

IMPACT OF EVOLUTION ON INFLUENZA A VIRUS FITNESS AND ANTIGENICITY DURING THE 2021- 2022 AND 2022-2023 FLU SEASONS

by

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Abstract

Influenza subtype dominance changes every season due to changes in antigenicity, fitness and even epidemiological factors allowing different subtypes to prevail during different seasons and geographical locations. The 2021-2022 and the 2022-2023 Influenza seasons were both H3N2 dominant seasons in the US. Sequences from circulating H3N2 viruses in the 2021-2022 flu seasons revealed significant genetic diversity leading to increased subclades undergoing parallel evolution. Fitness assays on representative viruses from Baltimore and Macha, including growth curves and plaque assays, revealed divergent plaque morphology. Phylogenetic analysis of viruses in the 2022-2023 influenza season revealed a possible reassortment event with mutations on important antigenic sites. No replication or plaque morphology changes were detected and circulating viruses were minimally drifted from the vaccine strain when tested against pre and post vaccination serum from healthcare workers. This indicates that these recent reassortment events and mutations have led to an expansion of genetic diversity to evolve from moving forward.

During 2022-2023 flu season the A5a.2 subclade was able to acquire several mutations within the HA segment that allowed it to bounce back and further diversify into A5a.2a and A5a.2a.1. These new subclades are marked by unique receptor binding and antigenic site mutations with A5a.2a.1 clade being additionally drifted from the vaccine. Representative viruses were characterized through plaque assays, low MOI growth curves and neutralization assays with serum from pre and post vaccinated healthcare workers. 2022-2023 circulating H1N1 viruses displayed similar or improved *in-vitro* fitness when compared to 2019 viruses. All 2022-2023 viruses were also antigenically drifted from the vaccine strain with A5a.2a.1 being additionally drifted. Single point

mutations on the HA segment were introduced on the A5a.2 backbone to interrogate their individual contributions. All mutations other than Q189E proved detrimental to the *in-vitro* fitness of these viruses. To investigate whether Q189E compensates for detrimental mutations plaque assays of these paired point mutants showed that while Q189E did improve fitness it did so in a reduced amount when combined with K142R. This data encapsulates the constant give and take between fitness and antigenicity.

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Table of Contents

Abstract	ii
Acknowledgements	iv
List of Tables	vii
List of Figures	viii
Chapter I: Introduction	1
1.1 Influenza: An Overview	2
1.2 Influenza: Evolution and Surveillance.....	3
Antigenic drift.....	3
Antigenic Shift	4
Importance of Surveillance	5
1.3 Influenza Replication.....	6
Step One: Viral Attachment	6
Step Two: Viral Entry.....	8
Step Three: Synthesis of Viral RNA.....	9
Step Four: Viral protein synthesis	10
Step Five: RNA packaging and Viral Assembly	10
Step Six: Budding and Release of Viruses	10
1.4 Antibody Responses to Influenza A Vaccinations.....	11
1.5 Overall Research Goal(s).....	14
Chapter II: General Methods	16
2.1 Ethics Statement and Human subjects.....	17
2.2 Cell cultures	17
2.3 Viral Isolation and Sequencing	18
2.4 Tissue culture infectious dose 50 (TCID ₅₀).....	19
2.5 Plaque Assay	20
2.6 Growth Curves	20
2.7 Serum Neutralization Assay	21
2.8 Statistical Analysis	22
Chapter III: The Evolution of H3N2 through the 2021-2022 and 2022-2023 Flu Seasons	23
3.1 Introduction	24
3.2 Results.....	25
3.2.1 2021-22 Local surveillance reflects globally circulating H3N2 clades and mutations, with Flu A activity registering lower levels compared to pre-pandemic	

years.	25
3.2.2 2021-22 Circulating H3N2 viruses from Baltimore and Macha did not show enhanced replication compared to the 2018-19 reference viruses but displayed divergent plaque morphology.	28
3.2.3 Local surveillance in 2022-23 reflects globally circulating H3N2 clades and mutations, with influenza activity spiking earlier and peaking higher than the previous season.	30
3.2.4 2022-23 H3N2 Influenza Season Sequencing and Phylogenetic Analysis Revealed a Possible Reassortment Event.	33
3.2.5 Circulating H3N2 viruses had similar replication and plaque morphology despite genetic diversity.	35
3.2.6 Circulating 2022-23 H3N2 viruses were minimally drifted from the vaccine strain.	37
3.3 Discussion and Conclusion	38
3.4 Methods	45
Chapter IV: H1N1 Evolution During the 2022-2023 Flu Season: The Battle between Fitness and Antigenic Drift.....	46
4.1 Introduction	47
4.2 Results	48
4.2.1 Evolutionary Dynamics of Influenza Subtype H1N1 during the 2022-2023 Season: Diversification of the A5a.2 Subclade.	48
4.2.2 Representative viruses from A5a, A5a.2, A5a.2a and A5a.2a.1 chosen for characterization.	49
4.2.3 2022-2023 Circulating H1N1 viruses display improved <i>in-vitro</i> fitness	51
4.2.4 All 2022-2023 circulating H1N1 viruses were antigenically drifted from the vaccine	53
4.2.5 Impact of A5a.2a and A5a.2a.1 mutations on <i>in-vitro</i> fitness.....	55
4.2.6 Q189E compensates for deleterious mutations.....	57
4.3 Discussion	58
4.4 Chapter Specific: Materials and Methods	62
Chapter V: References	66

List of Tables

Table 1: Neutralization Assay Serum Demographics	53
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List of Figures

Figure 1: Ribbon diagram of an uncleaved 1918 H1N1 Hemagglutinin monomer	8
Figure 3.1 Genetic diversity in H3N2 Circulating Clades Globally and Locally 2021-2022	27
Figure 3.2: In Vitro Viral Fitness of 2021-22 H3N2 Isolates.....	29
Figure 3.3: Genetic diversity in H3N2 Circulating clades Globally and Locally 2022-23 ..	32
Figure 3.4: Clade defining mutations and reassortment events in 2022-23 H3N2 viruses	34
Figure 3.5: In Vitro Viral Fitness of 2022-23 H3N2 Isolates.....	36
Figure 3.6: Serum neutralizing antibody titers in healthcare workers pre- and post-influenza vaccination.	38
Figure 4.1: Circulation and Emergence of the H1N1 A5a.2, A5a.2a & A5a.2a.1 clades.. ..	49
Figure 4.2: Genetic diversity in H1N1 Circulating clades Locally 2022-23.....	50
Figure 4.3: In Vitro Viral Fitness of 2022-23 H1N1 Isolates.....	52
Figure 4.4: Serum neutralizing antibody titers in healthcare workers pre- and post-influenza vaccination.	54
Figure 4.5: In Vitro Virus Fitness of A5a.2a & A5a.2a.1 Mutations in A5a.2 Backbone ..	56
Figure 4.6: Paired point mutant plaque assays. Quantitation of plaque areas of isolates	58

Chapter I: Introduction

1.1 Influenza: An Overview

Influenza viruses are enveloped negative-sense single-stranded RNA viruses with segmented genomes from the *Orthomyxoviridae* family. These viruses can be further classified into 4 types: Influenza A, B, C & D. Among these viruses, only influenza A and B have been implicated in human disease with influenza A being the only type to cause human pandemics. Influenza A viruses (IAVs) and Influenza B viruses (IBVs) are composed of 8 segments which are translated into both essential and accessory proteins that aid in the infection and replication of the virus. The 8 segments of IAV and IBV are PB1, PB2, PA, HA, NP, NA, M & NS. Influenza C and D viruses, on the other hand, are only composed of 7 segments due to the hemagglutinin-esterase fusion protein that replaces the hemagglutinin (HA) and neuraminidase (NA) proteins in IAVs and IBVs ¹⁻³.

When observed through electron microscopy IAV is generally spherical or elliptical in shape with some even being filamentous or irregularly shaped ⁴. The eight segments of influenza A and B are numbered from the largest to the smallest segments. In IAVs and IBVs the PB2, PA, HA & NP segments encode one protein per segment. Segment two of IAVs typically encodes PB1 with some strains also encoding the accessory protein PB1-F2. No similar protein has been found in IBVs. Segment six of the IAV encodes the NA protein and segment 7 encodes the M1 matrix protein. Finally, segment eight encodes for the interferon agonist NS1 and after splicing NS2 which has to do with viral ribonucleoprotein (RNP) export from the host's nucleus ⁵.

While sequencing has allowed us to confirm that the different Influenza viruses

share genetic ancestry, they have since evolved to the point where reassortment, involving the exchange of viral RNA segments between viruses, is only observed within each genus and type. Overall, Influenza viruses are characterized or grouped by their HA and NA proteins. Influenza nomenclature typically begins with virus type, species it was isolated from if not human, location, isolated number and in the case of IAVs, the HA and NA subtype (Ex: A/Darwin/9/2021). The HA and NA proteins are classified into subtypes, which were initially determined by the ability of antibodies to cross react with different virus isolates. Currently, the sequence of HA and NA proteins is used to classify them into subtypes. While there are many subtypes circulating the subtypes linked to epidemics have been either H1, H2 & H3 with either N1 or N2 ⁵ .

1.2 Influenza: Evolution and Surveillance

Antigenic drift

Influenza viruses acquire minor mutations in their HA and NA segments through a process known as antigenic drift ^{6,7}. This occurs due to the lack of proofreading mechanisms linked to the virally encoded RNA-dependent RNA polymerase. These point mutations occur randomly across the genome and could be either beneficial, detrimental, or neutral to virus replication with many mutations resulting in nonviable progeny. If a random mutation happens to occur in an antigenic site of the HA or NA glycoproteins, there is a potential for that virus to escape from preexisting immunity and therefore have a fitness advantage over other closely related viruses ^{6,8}. These mutations are then quickly selected for in the population because escape from preexisting immunity makes previously infected individuals

susceptible to infection with the new variant. The higher rate of mutations occurring in antigenic sites is attributed to the ability of these mutations to allow escape from preexisting host immunity ^{6,7}.

The ability of Influenza viruses to undergo antigenic drift is marked by the constant replacement of older variants with new, further drifted, variants. This could be further exacerbated by vaccination efforts that improve host immunity and potentially lead to a more rapid evolution. The process of antigenic drift allows the virus to escape antibody-mediated immunity and requires vaccines to be updated frequently. This further highlights the importance of surveillance programs to capture the antigenic changes that occur in the virus. This is typically done through the hemagglutination inhibition assays, neutralization assays or other similar assays⁷.

Antigenic Shift

Influenza viruses can undergo evolution in more ways than one. Apart from its ability to undergo point mutations through antigenic drift it can also undergo large antigenic changes through antigenic shift. The HA and in some cases the NA segments of this segmented virus can be swapped with other human or animal strains of the virus. The presence of 19 different subtypes of HA and 9 different subtypes of NA creates a pool of HA and NA proteins that are antigenically distinct from ones that are currently circulating in humans (H1-3 and N1-2). Humans have no preexisting immunity to H4-16 or N3-9 and therefore, an IAV encoding a combination of those HA and NA proteins could cause a pandemic. In some cases, animal influenza viruses can reassort with human viruses, facilitating the

emergence of an antigenically distinct virus that is capable of infecting and spreading in humans ^{7,9}. Antigenic shift has only been observed among IAVs due to the existence of animal reservoirs which can harbor antigenically distinct HA and NA subtypes. IBVs on the contrary do not have any known animal reservoirs and therefore have not been observed to undergo antigenic shift. Antigenic shift can result in a new variant that can infect a population that only has a limited level of immunity allowing for higher levels of transmission. All the human seasonal IAVs circulating today are descendants of pandemic strains that have coevolved with humans and drifted apart from their ancestors ⁶.

Importance of Surveillance

Influenza spreads globally in the form of seasonal epidemics and in some rare cases, pandemics as in the case 2009 H1N1 pandemic. Throughout history, we have seen several instances where influenza pandemic and epidemics have been devastating both in terms of health and economic impacts. Our discovery of vaccines has allowed us to come a long way in terms of controlling the spread and severity of influenza however, despite all efforts we have still not been able to eliminate this pathogen ^{10–13}. Influenza continually evolves through reassortment, and mutations leading to the constant rise and fall of new variants and subtypes ^{14–16}. The elimination of influenza is especially difficult due to the presence of multiple reservoirs (humans and animals in the case of IAV). Therefore, novel viruses can emerge and infect new hosts that may not have protection against it ^{17–19}. This means that the constant surveillance is important to control spread, assess the current impact and epidemiology of influenza and for the development of an

effective vaccination formulation and strategy.

1.3 Influenza Replication

Step One: Viral Attachment

Influenza viruses attach to cells through the N-acetylneuraminic acid (sialic acid) on the surface of host cells. These are nine-carbon monosaccharides that are typically present at the termini of different glycoconjugates. This means that they can be present on several different cell types in a range of animal species. The carbon-2 on the terminal sialic acid can bind to either carbon-3 or carbon-6 of galactose which results in an α -2,3- or α 2,6-linkage. The formation in these linkages can lead to distinctive steric configurations of the terminal sialic acid which is then identified and bound by HA proteins on influenza viruses which have a preference for these linkages. While both the α -2,3- and α 2,6 sialic acid receptors are present in humans, the human respiratory epithelium is predominantly occupied by α 2,6 linkages while the intestinal epithelium of avian species expresses α -2,3 linkages. These α -2,3 linkages are, however, present in the human respiratory epithelium meaning that humans and some primates can also be infected by avian viruses even if they are less efficient than the human strains. Additionally, humans have a higher number of α -2,3 linkages in the bronchioles and alveoli (lower respiratory tract). The varying levels of the different linkages can be attributed to the low infectivity and yet high pathogenicity of avian viruses in humans. Since the lower respiratory tract is not as accessible as the nasopharynx, paranasal sinuses, trachea, and bronchi it is not as common to see avian influenza infect humans, however, if they are able to successfully infect humans, it often

results in a severe and rapid pneumonia with high fatality rates ^{5,20,21}.

The crystal structure of the HA protein has revealed a trimer with two distinct regions: a globular head and a stem. The head of the protein contains the sialic acid receptor binding site which is in the middle of predicted variable antigenic determinants which have been labeled Sa, Sb, Ca1, Ca2 and Cb in the H1N1 subtypes and A, B, C and D in the H3N2 subtype **[Figure 1]** ⁵.

Over the course of virus replication, HA is cleaved into HA1 and HA2 by serine proteases which is required for viral infectivity. The HA protein is translated as a single polypeptide, which is inserted into the ER membrane and assembles into a trimer. This form of HA, called H₀, can bind to sialic acid receptor but cannot mediate membrane fusion. Cleavage of H₀ into HA1 and HA2 subunits converts the protein into a fusion competent form. The HA2 subunit is important for membrane fusion activity while HA1 encompasses the antigenic and receptor binding sites. Antibodies can prevent virus infection or neutralize infectivity by binding to specific regions of the HA protein, particularly regions close to the receptor binding region of HA. It is these antigenic sites that are subject to antigenic drift which allows the virus to evade immunity and make previously infected individuals susceptible to infection ^{5,22}.

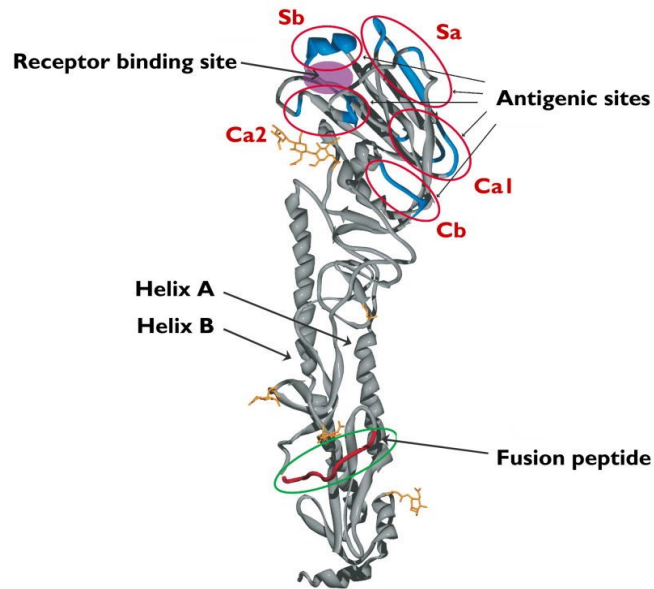


Figure 0-1: Ribbon diagram of an uncleaved 1918 H1N1 Hemagglutinin monomer. The receptor binding site is located on the head and is abutted the antigenic sites Sb, Sa, Ca1, Ca2 and Cb. The stem below is made up of alpha and beta helices as well as the fusion peptide⁵.

Step Two: Viral Entry

After the HA protein of the influenza virus attaches to the sialic acid linkage on host cells the virus is internalized through endocytosis. Once inside the endosome, two main processes come into play: First, the low pH initiates a conformational change in the HA protein which in turn exposes the fusion peptide leading to the merging of the viral envelope and the endosomal membrane resulting in the formation of a pore that allows viral RNP to enter the cell's cytoplasm. Second, the endosomal hydrogen ions are pumped into the viral M2 ion channel ^{5,23,24}, which allows the viral genome segments to be released from their tight association with the viral membrane and diffuse to the nucleus to initiate the next stage of the virus life cycle.

Step Three: Synthesis of Viral RNA

After RNPs leave the virion, they are trafficked into the host cell nucleus through viral proteins' nuclear localization signals (NLSs), which allow cellular proteins to transport RNPs and other viral proteins across the nuclear membrane²⁵. The nucleus along with an imported viral RNA-dependent RNA polymerase (RdRp) provides an environment to first synthesize the viral mRNA (capped, polyadenylated) that is recognized by the host and required to make viral proteins. RdRp uses the negative sense vRNA to synthesize 2 positive sense mRNA templates for the synthesis of viral proteins as well as the complementary RNA (cRNA) that allow the RNA polymerase to transcribe new copies of negative-sense genomic vRNA that is packaged in progeny virus particles ⁵.

The poly(A)tail of the Influenza virus mRNA is already encoded in its negative sense vRNA in the form of 5-7 uracil residues which is then transcribed into positive sense by viral polymerase that become a string of adenosines that eventually serve as the poly(A)tail. Another way the viruses disguises its RNA as host mRNA is by essentially using 5' capped primers derived from host pre-mRNA transcripts to prime the synthesis of viral mRNA through a process called "cap snatching"^{5,26,27}. After being polyadenylated and acquiring 5' caps, viral mRNA can then be exported and translated the same way as host mRNA.

vRNA segments on the other hand, are carried out of the nucleus through their association with the M1 and NEP/NS2 proteins. The M1 protein interacts with NP and the vRNA bringing the components together before associating with the nuclear export protein (NEP) which facilitates the export of M1-RNP through

nucleoporins to the cytoplasm ^{5,25}.

Step Four: Viral protein synthesis

The viral proteins HA, NA and M2 are made from viral mRNA on ER-membrane bound ribosomes that allow for the insertion of the proteins into the ER membrane. The proteins are then transported through the endoplasmic reticulum where they are properly folded before transport to the Golgi apparatus to undergo post-translational modification. These three proteins have apical sorting signals that lead them to the cell membrane where virion assembly can occur ⁵. The other viral proteins are cytosolic and are translated on free ribosomes in the cytoplasm, then transported to other sites within the cell.

Step Five: RNA packaging and Viral Assembly

While replication of genetic material and proteins are important, the virus must form an infectious particle to transfer its genetic material to a new cell. This involves the packaging of all eight viral genomic segments into one particle. This process was initially thought to occur at random, with only virions that happen to have all segments being capable of initiating a productive infection. However, new studies have found that packaging is a more selective process that involves discreet packaging signals on all segments increasing the odds of getting a full genome into more viruses ^{5,28,29}.

Step Six: Budding and Release of Viruses

It is hypothesized that the budding of influenza viruses at the cell's membrane is triggered by an accumulation of the M1 protein towards the cytoplasmic side of the

membrane. The accumulation of vRNPs at these sites of budding helps to trigger the initiation of budding. The HA and NA proteins cluster together at the plasma membrane and form “docking sites” for the M1 protein. This process leads to the concentration of viral proteins at budding sites. The receptor destroying activity of NA cleaves terminal sialic acid residues from the cell surface glycoproteins and the gangliosides to allow the emergence of progeny viruses. NA is also able to remove sialic acid residues from the virus particle to prevent the aggregation of virus particles. It also breaks down respiratory tract mucin secretions thereby allowing virus particles to move more efficiently from cell to cell. Antibodies targeting the NA protein or neuraminidase inhibitors can prevent the release of the virus from infected cells^{5,30}.

1.4 Antibody Responses to Influenza A Vaccinations

Influenza is a huge burden on human health through annual epidemics and less frequent pandemics ³¹. Therefore, vaccination and population immunity are paramount in limiting infections. Thanks to decades of research and the huge effort into developing vaccines, we have several different vaccines to help combat influenza viruses including, live attenuated virus vaccines (LAIVs), subunit vaccines, recombinant haemagglutinin vaccines, and whole virus vaccines. Each vaccine provides protection through different mechanisms ^{32,33}.

The goal of vaccination is to mimic infection through natural means, allowing us to generate a long lasting protective response against future infection ^{34,35}. However, the process of generating these vaccines whether it be through chicken eggs or cell culture can select for mutations that can interfere with the efficacy of these

vaccines ^{36–38}. Chicken eggs can result in egg-adapted mutations which affect the HA glycoproteins. While cell culture lowers the risk of egg adapted mutations, it is more expensive and the culture environment can lead to different mutations that affect viral antigenicity ^{39–41}.

Following vaccination, the viral RNA is detected first by the innate immune response followed by the induction of the effector and memory phases of the adaptive immune response. The humoral response comprises B cell mediated antibody responses. This response occurs swiftly upon antigen detection allowing for the production of antibodies that recognize viral proteins. B cell activation is key in generating vaccine-based protection ^{34,35}.

The 2022-23 seasonal inactivated influenza virus vaccines and LAIVs were quadrivalent consisting of two influenza A viruses and two influenza B viruses (A/Victoria/4897/2022 (H1N1) pdm09-like virus, A/Darwin/9/2021 (H3N2)-like virus, B/Austria/1359417/2021 (B/Victoria lineage)-like virus and a B/Phuket/3073/2013 (B/Yamagata lineage)-like virus) (CDC 2024). In contrast to other vaccinations against influenza, LAIVs are delivered in the nose, allowing it to replicate in the upper respiratory tract. This produces a local or mucosal immune response that protects from subsequent virus infection. Antibody responses to LAIV do not always involve increases in serum antibody levels. Studies conducted in cohorts or adults revealed seroconversion rates ranging from 3-13% for neutralization, 2-7% for hemagglutination inhibition, and 0-17% for NA inhibition with some studies even reporting significantly higher rates in all those readouts

9,43.

Subunit vaccines are produced by treating whole inactivated vaccines with detergent and then purifying the membranes that contain the HA and NA proteins. Based on how it is purified, this results in formulations that contain either components of HA and NA (in the case of a split virus vaccine) or just pure HA (in the case of a subunit vaccine). This vaccine is typically less reactogenic as a majority of the vRNA is purified out of the vaccine however this can also mean that the vaccine is less immunogenic. This type of HA targeted vaccination typically leads to hemagglutination inhibition or neutralizing antibody seroconversion rates of 20-77%. This type of vaccination can induce cross reactive antibody responses to older strains of the virus in adults that have preexisting immunity ^{9,44,45}.

Recombinant hemagglutinin vaccines only target HA but contain higher concentrations of HA than the split or subunit vaccines. The antibody responses to recombinant vaccines have been similar to levels associated with whole inactivated virus vaccines and either the split or subunit vaccines alike. There is also some evidence suggesting better responses and broader protection related to recombinant HA vaccines particularly in the elderly. These vaccines can also circumvent potential mismatches that can occur during passage in cell culture or eggs ^{9,46,47}.

Whole inactivated virus vaccines are well understood but rarely used due to their high reactogenicity. Certain inactivation methods allow antigens of live viruses to be more preserved and still have functioning HA and NA proteins. These inactivated vaccine viruses also have vRNA allowing for the activation of innate immune sensors such as retinoic acid inducible gene I as well as Toll-like receptors

(TLR) 3, 7 and 8 with the possibility of others allowing this vaccine to be self-adjuvating. This provides recipients with more holistic protections against HA and NA that can be observed in both humans and animal models. Seroconversion rates of 85% and above are also observed however, it is worth noting that some studies that performed head-to-head comparisons with split or subunit vaccines showed comparable rates of seroconversion (geometric mean hemagglutination inhibition assays) in humans with a study even showing better rates associated with subunit and split virus vaccine ^{9,48,49}.

1.5 Overall Research Goal(s)

With influenza constantly undergoing reassortment, antigenic drift and in some cases even antigenic shift, it is crucial to surveil and characterize all circulating viruses as well as assess current and post-vaccination population neutralization capacities. Apart from the rise and fall of variants, viruses also compete on a larger scale against other subtypes. The 2021-2022 and the 2022-2023 flu seasons were H3N2 dominant seasons. Sequences from circulating H3N2 viruses in the 2021-2022 flu seasons revealed significant genetic diversity leading to increased subclades undergoing parallel evolution. In the 2022-2023 seasons, H3N2 viruses went through significant reassortment with H1N1 viruses from the same season resulting in significant antigenic drift. Therefore, using samples collected through the Centers of Excellence for Influenza Research and Response (CEIRR) the aims of this project were to characterize circulating influenza A viruses during the 2021-2022 and 2022-2023 seasons, identify their ability to evade preexisting and vaccine induced immunity and define the contributions of important phenotype

defining mutations.

Chapter II: General Methods

2.1 Ethics Statement and Human subjects

The Serological samples used in this study were obtained from health care workers from the Johns Hopkins Centers of Excellence for Influenza Research and Response (JHCEIRRS) OHS Classic cohort. Serum was collected pre- and post-vaccination (~28 days) from all participants who provided written informed consent prior to participation. This recruitment was approved by the Johns Hopkins University School of Medicine Institutional Review Board, (IRB00288258). The viruses for this study were isolated from deidentified influenza A virus H3N2 positive samples, collected from consenting participants during the 2021-2022 and the 2022-2023 Influenza seasons at the Johns Hopkins Hospital, under the Johns Hopkins University School of Medicine Institutional Review Board, (IRB00091667).

2.2 Cell cultures

Three cell lines were used during this study: Madin-Darby canine kidney (MDCK) cells (provided by Dr. Robert A. Lamb), MDCK-SIAT cells (provided by Dr. Scott Hensley) and Human nasal epithelial cells (hNECs) (obtained from PromoCell, C-12620). MDCK and MDCK-SIAT cells were cultured in complete media (CM) prepared with Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, 100 units/ml penicillin/streptomycin (Life Technologies) and 2 mM Glutamax (Gibco) at 37 °C and 5% CO₂. hNECs were cultured in 24 well Corning transwell plates coated with 30 µg/ml rat tail collagen type I dissolved in PBS, and incubated at 37°C for at least 45 minutes and then rinsed with PBS. The cells were then incubated with PneumaCult Ex-Plus media (Stemcell #05040) + supplement on apical and basolateral sides. Once the cells grew to the

appropriate confluency (after approximately 10 days, decided through a trans-epithelial electrical reading of > 400 ohms), the medium was then changed to PneumaCult ALI medium and the apical surface was then exposed to air to form an air-liquid interface. Cells were incubated at 37°C and 5% CO₂. hNEC cells were considered fully differentiated based on the production of mucus and presence of ciliated epithelial cells.

2.3 Viral Isolation and Sequencing

Virus isolation was performed by adding nasopharyngeal swabs or nasal washes from influenza A H3N2 positive individuals during the 2021-2022 and the 2022-2023 Influenza seasons onto hNECs. The apical side of the hNECs were washed thrice with 150 µL of PBS, with 10-minute incubations in between each wash at 33°C and 5% CO₂. 150 µL of the sample was added to the apical side and incubated for 2 hours at 33°C and 5% CO₂. The inoculum was aspirated, and the apical side was washed twice with 150 µL of PBS. Virus was collected by adding 300 µL of infection media (DMEM supplemented with 0.3% BSA (Sigma), 100 units/ml pen/strep (Life Technologies), 2 mM Glutamax) to the apical side, incubating for 10 minutes, the supernatant was collected this way at days 3-, 5- and 7-days post infection. Viral isolation was confirmed and quantified using TCID₅₀ assays. Samples with a TCID₅₀/ml value greater than 10⁵ were selected to be made into seed stocks for use in subsequent experiments.

To prepare seed stocks, T75 Corning flasks are seeded with MDCK-SIAT cells and grown to full confluency in complete media. Viral isolates are then diluted to a multiplicity of infection (MOI) of 0.001 in infection media (infection media plus 5

µg/ml N-acetyl trypsin). Once confluent, the MDCK-SIAT monolayers were washed twice with PBS+ (PBS with 100 mg/L each of anhydrous CaCl₂ and MgCl₂.6H₂O) with the diluted viruses and incubated at 33°C and 5% CO₂ for 1 hour with gentle rocking every 15 minutes. The virus inoculum was then aspirated and the cells were washed once with PBS+ and 13 ml of infection media containing 5 µg/ml NAT was added to each flask and incubated at 33°C/ 5% CO₂. Once 75%- 80% of the cells in the flasks showed cytopathic effects (around 4 days post-infection), the medium was collected and centrifuged at 500g for 10 minutes at 8°C to remove cell debris. The supernatant was then aliquoted and stored at -80°C. Infectious virus concentrations in the seed-stock aliquots were quantified via TCID₅₀ assays.

Library preparation was performed using the Illumina RNA Prep with Enrichment(L) Tagmentation with Respiratory Virus Oligo Panel v2 (20044311) and viral RNA was extracted using QIAamp viral RNA mini extraction kit. The quality was checked using the Qubit and Agilent Bioanalyzer, and samples were sequenced using the MiSeq Illumina sequencer (v3 2X 300 bp). The DRAGEN RNA Pathogen detection pipeline was employed to generate consensus sequences using custom .bed files and FASTA files for IAV.

2.4 Tissue culture infectious dose 50 (TCID₅₀)

TCID₅₀ assays were performed by seeding 100 µL of MDCK-SIAT cells at a concentration of 1.5x10⁵ cells/ml in 96 well plates and allowing the cells to reach confluency in 48 hours. The 96 well plates were washed twice with PBS+ (PBS with 100 mg/L each of anhydrous CaCl₂ and MgCl₂.6H₂O) and filled with 180 µL of infection media supplemented with 5 µg/ml N-acetyl trypsin (NAT). Ten-fold serial

dilutions were performed on viruses and 20 μ L of the dilution was added to the corresponding wells in sextuplicate. The plates were incubated for 6 days at 33°C and 5% CO₂ and then fixed with 4% paraformaldehyde for > 3 hours and stained overnight with naphthol blue-black. The 50% tissue culture infectious dose was calculated as previously described^{50,51}.

2.5 Plaque Assay

Plaque assays were performed using MDCK cells using complete media on 6 well plates. Once the cells reached confluency, the complete media was aspirated and the cells were washed twice with PBS+. Ten-fold serial dilutions of the viruses were performed in infection media supplemented with NAT. A 250 μ L aliquot of each dilution was added to the wells and incubated at 33°C and 5% CO₂ for 1 hour with gentle rocking every 15 minutes to evenly distribute the plaques. After 1 hour, each well was overlaid with Temin's 2X MEM without phenol Red (Life Technologies) supplemented with 0.3% BSA, 2mM GlutaMax, 100 units/ml Penicillin/Streptomycin, 5mM HEPES buffer (Gibco), 5 μ g/ml N-acetyl trypsin, and 1% agarose. The plates were then incubated for 3 days at 33°C and 5% CO₂ and the plates were fixed with 4% formaldehyde in PBS overnight, the agarose overlay was removed and the cells stained using naphthol blue black the following day. Plaque areas were quantified using ImageJ.

2.6 Growth Curves

Growth curves were performed on fully differentiated hNECs at an air-liquid interface. The virus was diluted to an MOI of 0.01 in hNEC infection media to prepare the inoculum. The inoculum was then added to the cells and incubated for

2 hours after which the cells were washed 3 times with PBS and incubated with an air-liquid interface. At each timepoint 100 μ L of hNEC infection media was added to the wells and incubated at 33°C for 10 minutes. The media was then collected and quantified using the TCID50 assay.

2.7 Serum Neutralization Assay

Pre- and post-vaccination serum obtained through the Johns Hopkins Center for Excellence in Influenza Research and Surveillance study were used in this study. Prior to the neutralization assay, the serum samples were treated with Lyophilized Receptor Destroying Enzyme II (RDE, Hardy Diagnostics) dissolved into 20 ml of saline (0.9% NaCl in H₂O) and a 1:3 ratio of serum and RDE was incubated at 37°C overnight after which it was inactivated at 57°C for 35 minutes. MDCK-SIAT cells were seeded onto 96 well plates at a concentration of 1.5×10^5 cells/ml allowing the cells to reach confluency in 48 hours. Once the cells reached confluence, the plates were washed twice with 100 μ L per well of PBS+. The virus inoculum was prepared by diluting the virus stock in infection media to the concentration of 1000 TCID50/ml. The pre-RDE treated serum was then serially diluted two-fold in a 96 well U bottom dilution plate (thermo) using infection media and 110 μ L of virus was added to each dilution. This mixture was then incubated at room temperature for 1 hour and the 50 μ L of the mixture was transferred in quadruplicate to cell plates resulting in 25 TCID50 per well. After 24 hours, the mixture was removed, and the plates were washed once with PBS+ before adding 100 μ L of infection media supplemented with NAT. The plates were then incubated for 6 days at 33°C and 5% CO₂ and fixed with 4% formaldehyde for > 3 hours and

stained overnight with naphthol blue-black ^{15,52}.

2.8 Statistical Analysis

Statistical analyses was performed using GraphPad Prism 9.5. Plaque Assays were analyzed using 1- way ANOVA with Tukey Post-hoc test. The Growth curves were analyzed using 2- way ANOVA with Tukey post-hoc test. The Area under the curve (AUC) was calculated for each growth curve replicate and the values were then compared using a one-way ANOVA for differences in total virus production. All serology was analyzed through a 2-way ANOVA/1 way ANOVA, both or with the Tukey Post-hoc test.

Chapter II: The Evolution of H3N2 through the 2021-2022 and 2022- 2023 Flu Seasons

3.1 Introduction

The year 2023 marked the 55th anniversary of the 1968 Influenza A H3N2 pandemic ⁵³. Influenza is a seasonal fixture in the human population. Its resilience lies in its unique ability to undergo antigenic drift. Accumulation of random mutations in the viral genome can impact viral fitness, receptor binding or the ability of the virus to evade existing immunity ^{54,55}. Consequently, new clades can emerge, and reinfect individuals who were previously infected or vaccinated. Despite extensive vaccine efforts and research, influenza remains a significant cause of morbidity and mortality annually, highlighting the virus's enduring impact.

During the 2021-2022 influenza season, there was a rise in cases that started in November 2021, declined in January 2022, and then rose again in March 2022 staying high until June 2022. There were two distinct waves that peaked first towards the end of December 2021 and second in May 2022. The United States had an estimated 9,297,886 symptomatic cases with 4,314,281 medical visits and 101,262 hospitalizations with 4,601 deaths caused by influenza. The highest burden was found to be present in the population above 65 years. This season was described to be of low severity (CDC, 2023). In the prior influenza season, 2020-2021, influenza activity was extremely low with only 0.2% of the respiratory specimens testing positive. This low influenza rate was attributed to the public health measures taken to combat the COVID-19 pandemic. This resulted in influenza and other respiratory viruses, going through unique genetic bottlenecks (CDC, 2022). While the 2021-2022 season did see more activity than the 2020-2021 season, it was not as high as previous seasons.

While the 2022-2023 influenza season also began in October, we saw an unusual and rapid rise in influenza cases that peaked November to early December 2022 rather than the typical December to early January peak. There were an estimated 26 million illnesses with 290,000 hospitalizations and 18,000 deaths in the United States (Grohskopf et al., 2024; CDC, 2024). The 2022-2023 Influenza season was a predominantly H3N2 season with this type making up 70.8% of the cases as of August 2023. The season began in October 2022 there was an early and unprecedented surge of cases that started in November 2021 and remained high all the way until mid-June making it active later than any other flu season since the 2009 H1N1 pandemic (CDC, 2024).

Vaccination has been our greatest asset in preventing the severity and reducing cases of influenza. However, the COVID-19 pandemic has affected the pattern of other seasonally circulating viruses such as influenza. This has been attributed to the implementation of non-pharmaceutical interventions limiting the circulation of many pathogens. The combination of these unique circumstances and the rapidly evolving nature of influenza makes it especially important to understand and characterize seasonal viruses such as influenza to inform vaccine design more accurately ^{54,60–62}.

3.2 Results

3.2.1 2021-2022 Local surveillance reflects globally circulating H3N2 clades and mutations, with Flu A activity registering lower levels compared to pre-pandemic years.

Phylogenetic and sequence analysis was performed on 278 H3N2 influenza A

virus samples that possessed complete HA sequences revealing that viruses circulating during the 2021-2022 season included viruses from the subclade 3C.2a1b.2a.2a as well as viruses that emerged from this subclade. Circulating H3N2 viruses predominantly belonged to the 3C.2a1b.2a.2a.1a (64.39%) followed by 3C.2a1b.2a.2a.3 (19.42%), 3C.2a1b.2a.2a.1 (9.71%) and 3C.2a1b.2a.2a (6.47%) **[Figure 3.1A & B]**. Flu A activity was higher than the 2020-2021 influenza season however, still lower than pre-pandemic years **[Figure 3.1C]**.

A total of 58 nasal swabs obtained from influenza positive patients through the Clinical Virology laboratory of the Johns Hopkins Hospital Network were inoculated on differentiated hNECs. Out of 58 samples, 32 samples were influenza A virus samples and all the H3N2 viruses belonged to the 3C.2a1b.2a.2a.1a subclade.

17 nasal swabs enrolled from the Macha, Zambia, 12 from H3N2 and 4 being flu negative, were all inoculated on differentiated hNECs out of which 2 successful isolates from the dominant clade 3C.2a1b.1a were obtained.

The two chosen representative viruses circulating in Baltimore were A/Baltimore/JH-020/2021 and A/Baltimore/JH006/2021 from the clade 3C.2a1b.2a.2a.1a and the two viruses chosen to represent the dominant clade 3C.2a1b.1a in the Macha region of Zambia were A/Zambia/O1237/2021 and A/Zambia/O1252/2021. All chosen viruses were inoculated on MDCK-SIAT1 cells to generate seed stocks for viral characterization.

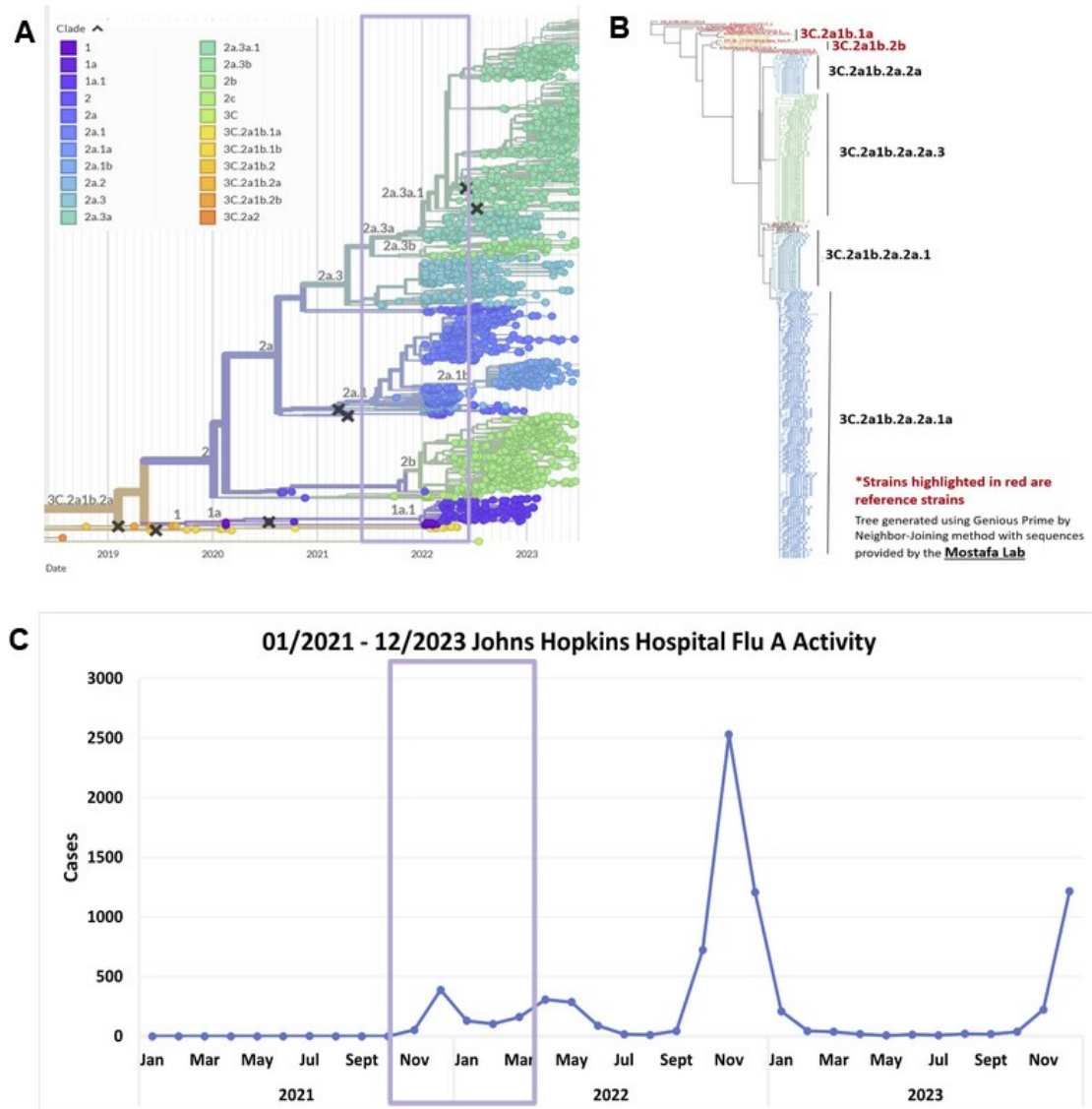


Figure 3.1 Genetic diversity in H3N2 Circulating Clades Globally and Locally 2021-2022. A) Phylogenetic Tree showing the sequence of HA genes of H3N2 viruses from 2019-2023 with clades circulating in North America and Africa highlighted. Generated through the NextStrain pipeline. X indicates vaccine strains. The box indicates the 2021-2022 Northern Hemisphere Influenza Season. B) HA phylogenetic tree showing the different H3 HA clades circulating in Baltimore and in Macha during 2021-2022. 2018-2019 viruses from 3C.2a1b.2b have been included to indicate genetic differences. Reference strains are in red. Constructed using Neighbor-Joining method, Tamura-Nei distance model. C) Graph showing FluA activity across Jan/2021 to Dec/2023 in the Johns Hopkins Hospital with the 2021-2022 season indicated.

3.2.2 2021-2022 Circulating H3N2 viruses from Baltimore and Macha did not show enhanced replication compared to the 2018-19 reference viruses but displayed divergent plaque morphology.

To access replication differences, growth curves were performed on a physiologically relevant cell culture model, human nasal epithelial cell (hNEC) cultures. Infecting cells at a MOI of 0.01 revealed no significant replication differences between all 6 H3N2 clades **[Figure 3.2A]**. To further assess differences in the *in-vitro* fitness of these isolates, plaque assays were performed on MDCK cells to identify differences in size and morphology. Circulating early season Baltimore H3N2 viruses (3C.2a1b.2a.2a.1a) produced significantly larger plaques than the Baltimore and Macha 2018-19 reference viruses (3C.2a1b.2b). The two representative 3C.2a1b.1a isolates from Macha produced different sized plaques, with 01252 producing similar plaques to the 2018-19 reference viruses, while the 01237 strain produced significantly smaller plaques than the Macha reference strains. Finally, 2021-22 circulating Baltimore viruses produced significantly larger plaques than both the circulating and reference isolates from Macha **[Figures 3.2B & C]**. Taken together, the data indicate that there were no significant replication differences in hNEC cultures between all H4N2 viruses tested, though there were some differences in plaque size in MDCK cells.

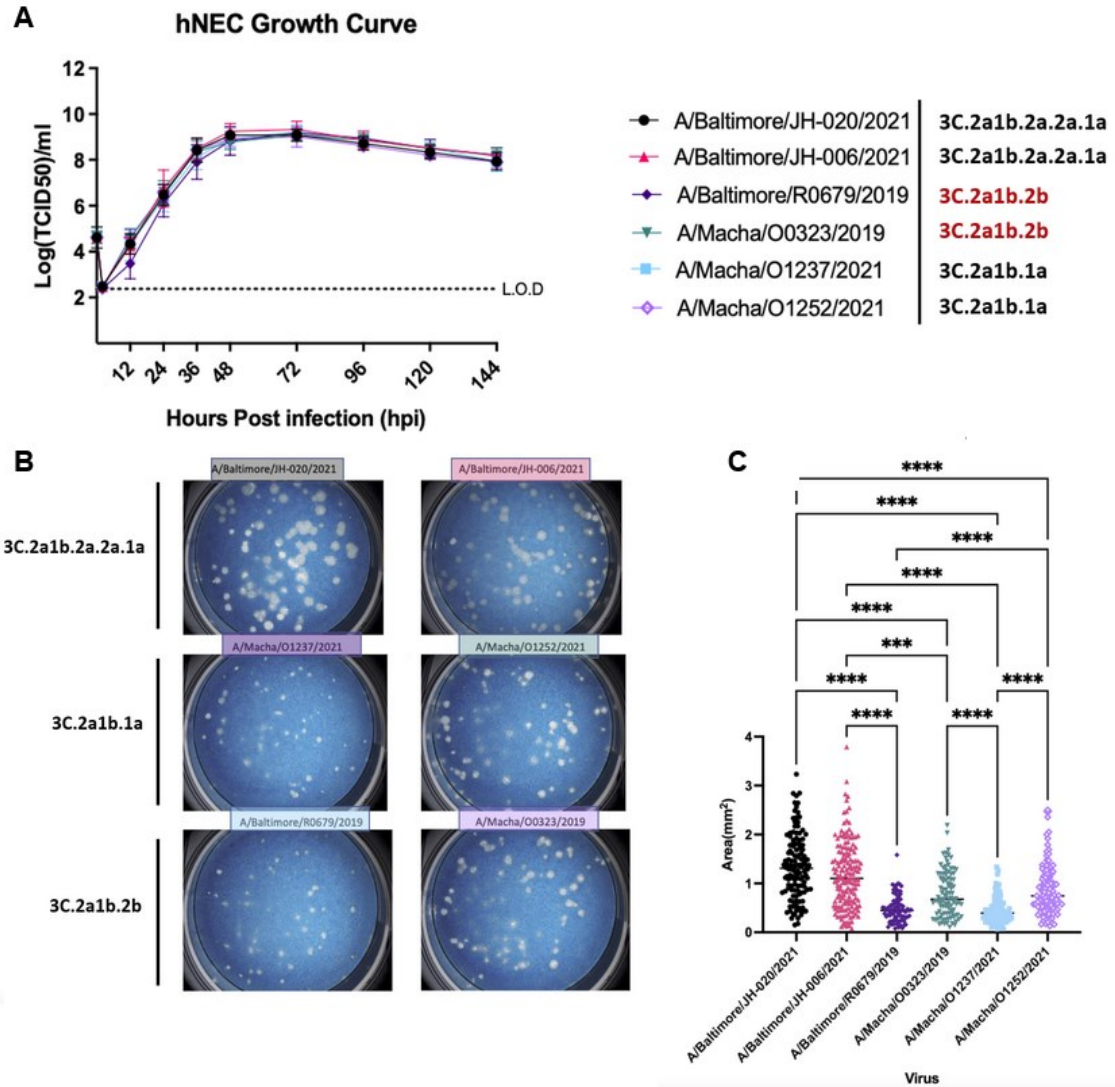


Figure 3.2: In Vitro Viral Fitness of 2021-22 H3N2 Isolates. A) Infectious virus production after infection of hNEC cultures. Data analyzed by two-way ANOVA with Tukey post hoc test. Each graph combines three experiments with four replicates per isolate per experiment. B) Representative images of plaques in MDCK cells. C) Quantitation of plaque areas of isolates. Three independent assays with >50 plaques quantified per isolate. One-Way Anova with Tukey post-hoc test, *** $p < 0.001$, **** $p < 0.0001$.

3.2.3 Local surveillance in 2022-2023 reflects globally circulating H3N2 clades and mutations, with influenza activity spiking earlier and peaking higher than the previous season.

Phylogenetic and sequence analysis was performed on 2022-2023 H3N2 samples that possessed complete HA sequences. HA phylogenetic analysis revealed expanded diversity, resulting in new, genetically distinct subclades that are co-circulating and undergoing parallel and diversifying evolution. These distinct clusters have been provided with new clade designations: 3C.2a1b.2a.2a, 3C.2a1b.2a.2b, 3C.2a1b.2a.2c and 3C.2a1b.2a.2d. All these subclades possess the clade defining mutations of the 3C.2a1b.2a.2 clade with the addition of additional unique HA mutations [**Figure 3A & B**]. Flu A activity was substantially higher than the 2021-2022 influenza season and started earlier (Sept) [**Figure 3.3C**].

Phylogenetic analysis of samples obtained through the Clinical Virology Laboratory of the Johns Hopkins Hospital network in Baltimore revealed that the five viruses in circulation were 3C.2a1b.2a.2b (64.14%), 3C.2a1b.2a.2a.3a.1 (10.86%), 3C.2a1b.2a.2a.1 (9.43%), 3C.2a1b.2a.3 (1.43%), 3C.2a1b.2a.2a.1b (6.97%) and 3C.2a1b.2a.2b (6.97%). Further phylogenetic analysis of the other seven genetic segments (NA, PA, PB1, PB2, M, NP, and NS) of these viruses revealed that a number of the 2022-2023 H3N2 viruses in circulation had 6-7 segments with sequence similarity to the segments in the 3C.2a1b.1a clade indicating a potential reassortment event ⁶³.

To characterize the evolution of these viruses, 56 chosen clinical samples,

obtained from the Johns Hopkins Hospital network, were inoculated on differentiated hNECs. Of the 26 viruses that were successfully isolated, one virus was chosen from each subclade.

The viruses chosen for characterization were A/Baltimore/JH0380/2022 (3C.2a1b.2a.2a.1) with 6 reassorted segments, A/Baltimore/JH-0442/2022 (3C.2a1b.2a.2a.1b), A/Baltimore/JH0339/2022 (3C.2a1b.2a.2a.3) with 6 reassorted segments, A/Baltimore/JH-0586/2022 (3C.2a1b.2a.2a.3a.1), A/Baltimore/JH-0515/2022 (3C.2a1b.2a.2b) and A/Baltimore/JH0440/2022 (3C.2a1b.2a.2b).

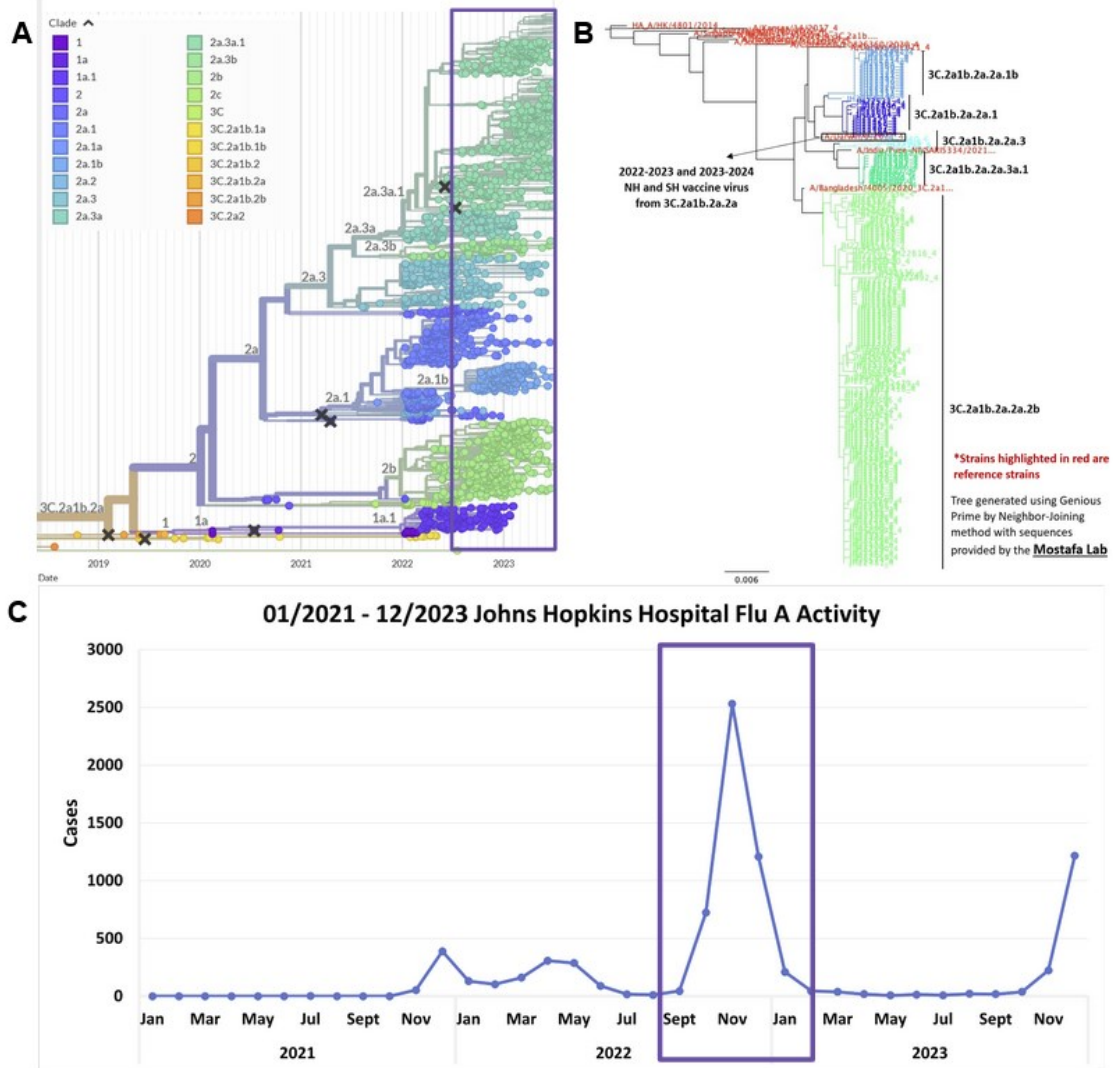


Figure 3.3: Genetic diversity in H3N2 Circulating clades Globally and Locally 2022-2023. A) Phylogenetic Tree showing the sequence of HA genes of H3N2 viruses from 2019-2023 with clades circulating in North America and Africa highlighted. Generated through the NextStrain pipeline. X indicates vaccine strains. The box indicates the 2022-23 Northern Hemisphere Influenza season. B) HA phylogenetic tree showing the different clades circulating in Baltimore and Zambia during 2022-2023. 2018-2019 viruses from 3C.2a1b.1a & 3C.2a1b.2b have been included to indicate genetic differences. 2022-2024 Northern hemisphere and Southern hemisphere vaccine virus from 3C.2a1b.2a.2a also indicated. Reference strains are in red. Constructed using the Neighbor-Joining method, Tamura-Nei distance model. C) Graph showing FluA activity across Jan/2021 to Dec/2023 in the Johns Hopkins Hospital with the 2022-23 season indicated.

3.2.4 2022-2023 H3N2 Influenza Season Sequencing and Phylogenetic Analysis Revealed a Possible Reassortment Event.

Phylogenetic Analysis of the 2022-23 H3N2 Influenza season unveiled a possible reassortment event. Looking further into the clade defining mutations show that they have been accrued on important antigenic sites **[Figure 3.4A]**.

Looking further into the sequences showed that there was a possible reassortment between 3C.2a1b.2a.2b and 3C.2a1b.1a resulting in the subclades 3C.2a1b.2a.2b with only either HA and NA from the 3C.2a1b.2a.2b or just the HA from that clade. The 3C.2a1b.1a clade also reassorted with the 3C.2a1b.2a.2a to 3C.2a1b.2a.2a with all segments other than HA and NA being from 3C.2a1b.1a. The unassorted 3C.2a1b.2a.2a clade accrued further mutations and launched the subclades 3C.2a1b.2a.2a.1b and 3C.2a1b.2a.2a.3a.1 while the 3C.2a1b.2a.2a reassorted clade acquired further mutations leading to the subclade 3C.2a1b.2a.2a.3 **[Figure 3.4B]**.

3.2.5 Circulating H3N2 viruses had similar replication and plaque morphology despite genetic diversity.

Replication differences were accessed using growth curves performed on hNEC cultures. Infection with a low MOI of 0.01 revealed no significant replication differences between all 6 circulating viruses [**Figure 5A**]. Plaque assays were performed on MDCK cells and no significant differences in plaque sizes or morphology were detected [**Figure 3.5B & C**].

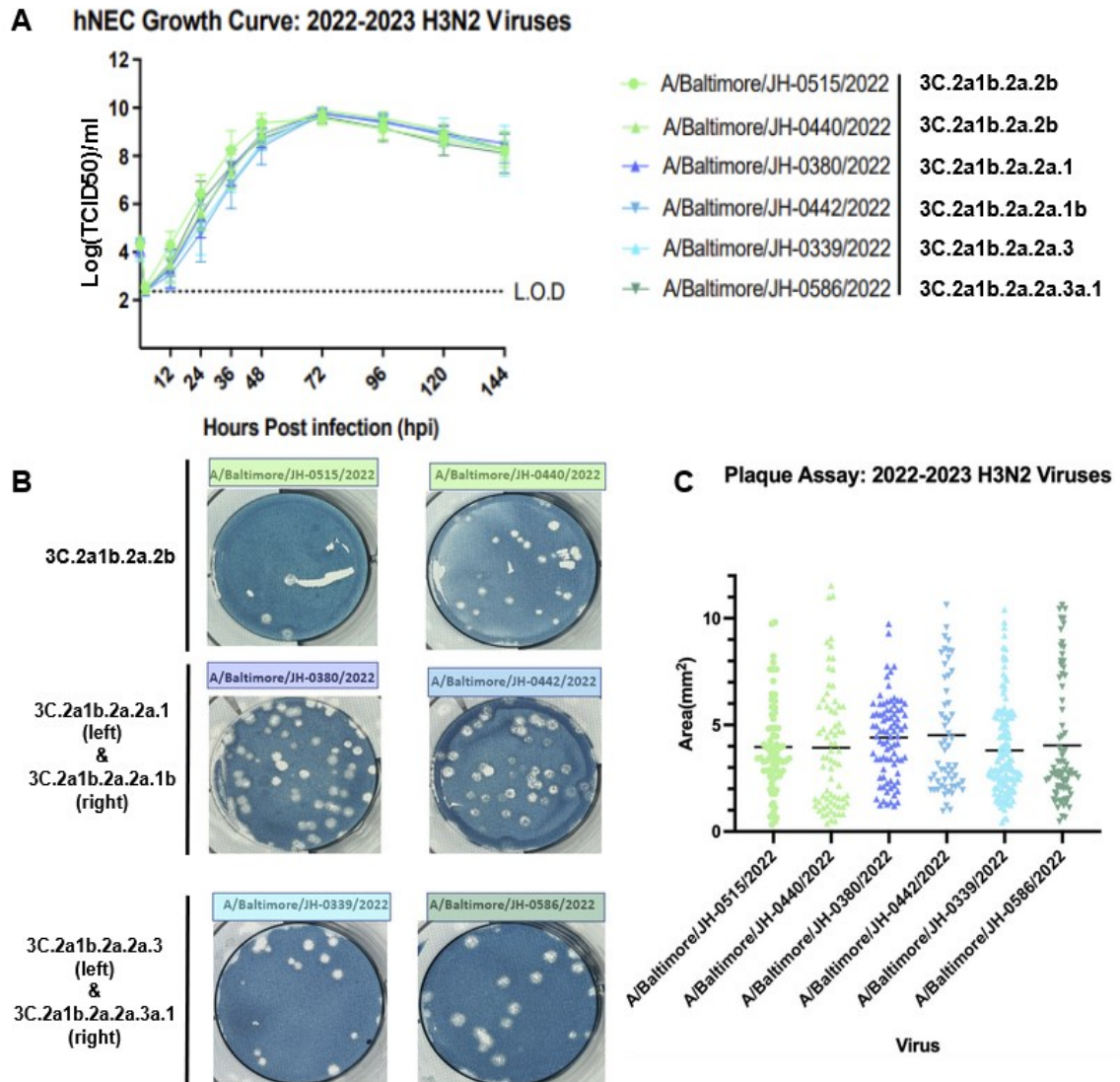


Figure 3.5: In Vitro Viral Fitness of 2022-2023 H3N2 Isolates. A) Infectious virus production after infection of hNEC cultures. Data analyzed by two-way ANOVA with Tukey post hoc test. Each graph combines three experiments with four replicates per isolate per experiment. B) Representative images of plaques in MDCK cells. C) Quantitation of plaque areas of isolates. Three independent assays with >50 plaques quantified per isolate. One-Way Anova with Tukey post-hoc test, *** $p < .0001$.

3.2.6 Circulating 2022-23 H3N2 viruses were minimally drifted from the vaccine strain.

While growth curves and plaque assays provide valuable information regarding the replication fitness of a virus, it is also important to assess the ability to evade vaccine induced or pre-existing immunity in the human population. Antigenic escape was judged by neutralization assays performed with serum acquired from healthcare workers before and after receiving their annual influenza vaccination for the NH 2022-23 season **[Table 1]**.

Representative viruses were chosen from the 2a.1 (JH-0380), 2b (JH-0515) and 2a.3a.1 (JH-0586) and compared to the vaccine strain (A/Darwin/9/2021). In pre-vaccination serum, the only circulating clade to show reduced serum neutralization was 2a.1. Post-vaccination, only the 2b clade showed reduced serum neutralization when compared to the vaccine strain. **[Figure 3.6A]**. The fold change before and after vaccination was similar for the vaccine and circulating clades, indicating that vaccination induced antibody responses against all the tested viruses with approximately the same efficiency **[Figure 3.6B]**.

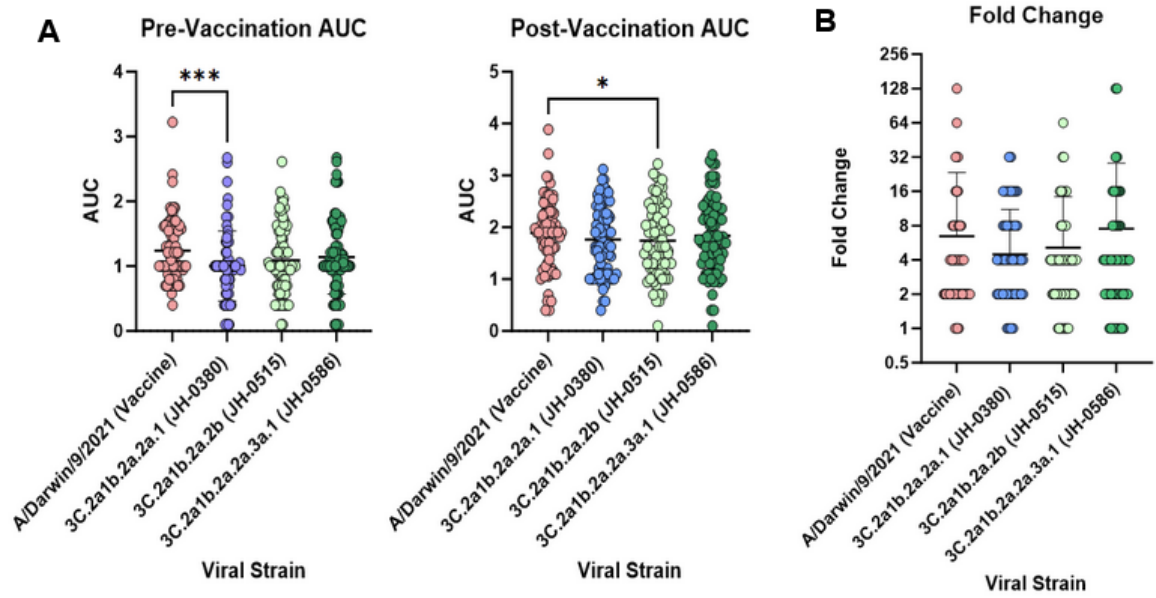


Figure 3.6: Serum neutralizing antibody titers in healthcare workers pre- and post-influenza vaccination. A) Serum neutralization titers against the H3N2 vaccine strain (A/Darwin/9/2021), 3C.2a1b.2a.2a.1 (A/Baltimore/JH-0380/2022), 3C.2a1b.2a.2b (A/Baltimore/JH-0515/2022), 3C.2a1b.2a.2a.2a.1 (A/Baltimore/JH-0586/2022). Area under the curve (AUC) was calculated and analyzed using 1-way ANOVA with Tukey post-hoc, * $p < 0.05$, * $p < 0.001$. B) The fold-change of serum neutralizing antibodies compared across vaccines and circulating isolated. Bars represent geometric mean, intervals are geometric standard deviations. One-way ANOVA with Tukey post-hoc. N=77.**

3.3 Discussion and Conclusion

Seasonally circulating influenza viruses infect around 3-11% of the US population and around 5-15% of the global population, every year (CDC, 2022; Petrova & Russell, 2017). This annual burden is credited to the ability of the virus to undergo antigenic shift and drift allowing them to escape immunity gained through vaccination or infection with prior strains of the virus. While escaping immunity grants these viruses a greater chance of infection, some mutations can also improve the fitness of a virus such as its ability to spread from cell-to-cell, bind to different types of cells, replicate, and even transmit from person-to-person.

Therefore, influenza vaccines need to be continually updated to protect the population from currently circulating viruses ^{54,64,66}.

When COVID-19 went from an outbreak to a pandemic on March 11, 2020, countries all around the world were strongly encouraged to deploy strict non-pharmaceutical interventions (NPIs) such as social distancing, masks, and quarantine to protect the public ⁶⁷. While these interventions helped curb the spread of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), it also impacted the transmission of other respiratory viruses such as influenza which experienced lower levels of activity during that year and seasons that followed. However, the relaxation of NPIs allowed a more susceptible population to be exposed to seasonal viruses resulting in more infections and healthcare-seeking rates around the world when compared to before the pandemic. This could indicate that there might be higher influenza activity if immunity and NPIs are reduced. Additionally, this may indicate that long term NPIs such as masks and social distancing may have an impact on immunity ^{61,68}.

Influenza viruses have always evolved under high positive selection pressure due to its need to escape preexisting immunity forcing it to constantly evolve through antigenic drift and genetic selection. Overall, the combination of the virus's typical evolution, the decline of immunity in the population and the new proposed burden of 'immune debt' could mean reestablishing our balance with the virus ^{61,68}. Therefore, examining the influenza viruses circulating in the 2021-23 seasons will allow us to study the impact of the COVID-19 pandemic on influenza evolution.

The phylogenetic analysis of whole viral genome sequences from viruses circulating in Baltimore in the 2021-2022 influenza season predominantly belonged to the 3C.2a1b.2a.2 clade. These viruses emerged from the 3C.2a1b viruses that were the predominant virus in circulation between September 2018 to January 2019 (Melidou et al., 2020; WHO, 2019) [**Figure 3.1A**]. While the 3C.2a1b.2a.2 viruses predominated in the United States and several other parts of the world, viruses circulating in Macha, Zambia early June 2021 belonged to the clade 3C.2a1b.1a. Taken together, viruses predominantly circulating in the US during the 2021-2022 influenza season were 3C.2a1b.2a.2 with 3C.2a1b.2a.1, 3C.2a1b.1a, and 3C.2a1b.1b circulating at lower frequencies. The H3N2 component of the vaccine formulation deployed to circumvent public health burden in this year belonged to the 3C.2a1b.2a.1a clade. While the ferret antisera generated against the egg-based vaccine virus was able to effectively neutralize circulating viruses in the same clade, the antiserum was not as effective against all of the other co-circulating clades (WHO, 2019). The low seroconversion rate after vaccination combined with reduced exposure to circulating H3N2 viruses during the COVID-19 pandemic could have contributed to the transmission and evolution of 3C.2a1b.2a.2 viruses.

Using the 3C.2a1b.2b clade as a reference (A/Baltimore/R0679/2019 and A/Macha/O0323/2019), we studied the evolution and circulation of viruses that initially evolved from the subclade 3C.2a1b into 2 distinct clades, 3C.2a1b.2a (A/Baltimore/JH020/2021 and A/Baltimore/JH-006/2021) and 3C.2a1b.1 (A/Macha/O1237/2021 and A/Macha/O1252/2021) in the 2021-22 influenza

season while other clades seemed to have gone extinct. The overlap in regions relating to fitness and immune escape could cause mutations in the genome to improve one trait while compromising another. Therefore, the prevalence of one virus over the other could be due to fitness and/or antigenic escape advantages possibly selected for by the unique pandemic bottleneck ^{15,71,72} can be parsed out by characterizing the *in vitro* fitness of these viruses through hNEC growth curves and MDCK plaque assays.

Interestingly, while there were no differences in replication, there were significant differences in plaque morphology. The plaque sizes of all the H3N2 viruses circulating in Baltimore during the 2021-2022 (3C.2a1b.2a.2a.1) influenza season were larger than the 2018-2019 3C.2a1b.2b viruses circulating in both Baltimore and Macha. Despite both viruses from Macha being from the 3C.2a.1b.1a clade, one strain produced plaques smaller than the reference virus while the other one produced similar sized plaques. This illuminates the variability in plaque morphology within one clade [Figure 3.2]. This data also indicated that the 3C.2a1b.2b clade from 2019 possessed mutations that improved the ability of viruses to spread from cell-to-cell that the 3C.2a1b.1 did not gain. While we do not know the exact differences in receptor binding, using the physiologically relevant hNEC model that has an array of α 2-6 and α 2-3 linked glycans capturing physiologically relevant conditions in the human respiratory tract, showed no differences between any of the 2021-2022 isolates. Since hNEC cultures are a more physiologically relevant cell culture system, we would conclude that these viruses all had equivalent replication potential, thereby allowing all of them to

circulate in the human population. In studies of seasonal H1N1 viruses, we did see a close correlation between replication in hNEC cultures and MDCK plaque size so the inconsistency may be specific to H3N2 viruses. It is important to note that while we characterize the isolates based on their HA clade designation, there are genetic differences in other gene segments that could be contributing to the differences in MDCK plaque morphology. ^{3,22–24}

The 2022-2023 influenza season saw a more severe and early peak (late November - early December) and a rapid decrease in cases (CDC, 2024) [**Figure 3.3C**]. To elucidate the virological contribution to these differences, viruses from this season were characterized through *in vitro* fitness assays and their ability to evade vaccine-induced and preexisting immunity with serum neutralization assays. While point mutations can lead to antigenic drift, reassortment of segments can lead to improved virus fitness and given the diversity of HA clades present in this year, we were interested in seeing how antigenic structure and fitness were affected. ⁵³.

Phylogenetic analysis of samples from the 2022-2023 season revealed that circulating H3N2 viruses not only underwent genetic drift – as judged by the diverse HA clades in circulation - but also reassortment between circulating H3N2 clades [**Figure 3.3 & 3.4**]. Reassortment between clade 3C.2a1b.1a and the subclades 3C2a1b.2a.2b led to two distinct reassorted viruses that differed by the NA segment. Reassortment also occurred between clades 3C.2a1b.1a and 3C.2a1b.2a.2a, generating additional genetic changes in HA that resulted in this reassortment being characterized as clade 3C.2a1b.2a.2a.1. This clade further

evolved through point mutations to result in clade 3C.2a1b.2a.2a.3. In addition to these reassortment events, the 3C.2a1b.2a.2a clade viruses that did not undergo reassortment acquired additional mutations in HA, leading to the two subclades: 3C2a1b.2a.2a.1b and 3C.2a1b.2a.2a.3a.1 **[Figure 3.3A & B]**.

Despite the accumulation of HA mutations and reassortment, the *in vitro* fitness of circulating 2022-23 H3N2 viruses in Baltimore revealed no replication differences in hNECs or disparate plaque morphologies, suggesting that all the isolates were equally fit by these parameters **[Figure 3.5]**. Every segment of the virus is crucial to the production of viable progeny therefore, while some mutations or reassortment events are beneficial, some can be detrimental ^{15,74}. The evolution of influenza viruses produces a continuously evolving lineage with proteins that all depend on each other. This balance is significantly impacted by progeny with a whole new combination of segments which can result in a fitness cost that could be balanced out with genetic diversity. Intra-lineage reassortment is faced with considerable negative selection which strengthens as the distance between the circulating and parental strains grow ⁷⁵. It remains to be seen if the reassortment events conferred some other advantage to those viruses, increased transmission efficiency for example, which were not measured by our assays.

Several mutations accrued by these viruses are located on antigenic and receptor binding sites **[Figure 3.4A]**. These mutations might induce antigenic changes compared to the 3C.2a1b.2a.1 (A/Cambodia/E0826360) vaccine strain. Vaccine efficacy was shown to be higher in the 2022-2023 flu season when compared to the 2021-2022 season ^{76–78}. Serum collected from a cohort of healthcare workers

vaccinated in 2022-2023 showed no clear antigenic drift with any of the circulating HA clades. This suggests that no clade had a clear advantage in evading preexisting or population immunity. We hypothesize that the 2022-2023 season represented a year where genetic diversity led to the evolution of several different HA clades and reassorted viruses of equal ability to circulate in the population. With no clade having a clear advantage in the human population, this genetic diversity was conserved throughout the influenza season.

While it is tempting to assume that the reassortment events are what led to the more severe influenza season we observed in 2022-2023 [**Figure 3.3A & 3.4**], there are a combination of factors at play this year. The early and steeper peak could be explained by a range of other factors such as the decrease in herd immunity to seasonal influenza, the ease of NPIs against COVID-19, the return in the typically transmission pattern of influenza, and even the emergence of a strain more fit and able to escape immunity ⁶³. Why H1N1 or influenza B viruses did not circulate at high levels during this time is not clear, as the same factors that increased susceptibility to H3N2 infection should have also led to increased H1N1 and influenza B virus circulation.

Reassortment has been shown to be a critical factor allowing influenza viruses to evolve faster and to develop an enhanced ability to jump from one species to another potentially giving rise to pandemics ⁷⁹. The reassortment of HA and NA proteins particularly, has been linked to both the ability to escape immunity and functional epistasis, both of which affects the vaccine efficacy for that year ⁷⁵. Unlike in the case of the 2022-2023 H3N2 reassortment event, there are several

examples of reassortment events contributing positively to the fitness of a virus specifically in the case of H3N2⁸⁰. One such example is when reassortant viruses were generated between cocirculating H5N1 and human H3N2 which resulted in H5N1 reassortant viruses with H3N2 PB2 increasing the viability and replication of the virus⁸¹. Chen et al. (⁸²) also demonstrated an increase of virulence following reassortment between human isolated of H3N2 and H5N1 NA of the H3N2 virus in the H5N1 background. The similar replication and plaque morphology of these viruses all indicate a potential stochastic event setting the stage for more favorable mutations moving forward.

3.4 Chapter Specific Methods

3.4.1 Ethics Statement and Human subjects, Cell culture, Viral Isolation and Sequencing, TCID50, Plaque Assay, Growth Curve, Serum Neutralization Assay, Statistical Analysis – As described in Chapter II

3.4.2 Viruses used:

- A/Baltimore/JH-020/2021
- A/Baltimore/JH006/2021
- A/Zambia/O1237/2021
- A/Zambia/O1252/2021
- A/Baltimore/JH0380/2022
- A/Baltimore/JH-0442/2022
- A/Baltimore/JH0339/2022
- A/Baltimore/JH-0586/2022
- A/Baltimore/JH-0515/2022
- A/Baltimore/JH0440/2022

Chapter IV: H1N1 Evolution During the 2022-2023 Flu Season: The Battle between Fitness and Antigenic Drift

4.1 Introduction

Human influenza viruses are genetically dynamic resulting in different types, subtypes, and clades. This extant and potential diversity makes it important to constantly surveil the human population, identify which subtype is dominant and adjust vaccination policies and formulations accordingly ^{54,64,66}. The 2022-2023 season was a moderate Influenza season that was closer to pre-COVID-19 levels of activity (CDC, 2023). What was particularly interesting about this season was that it started earlier than previous years and that this was a particularly severe season for children ⁸⁴. This season was a stark reminder of both the burden of influenza on vulnerable populations such as children and the elderly but also the importance of prevention measures against this disease ⁸⁴.

Human influenza subtypes go through antigenic drift every year, driven by random mutations in the viral surface proteins. These mutations can affect either the fitness of the virus or its ability to evade the immune system. Viral fitness refers to the ability of the virus to generate infectious progeny in a specific environmental setting. Immune pressure and host mortality means that for a virus to survive, they need to transmit. How well they evade protective immune mechanisms and transmit defines which variant is dominant in the environment. It has evolved that neutralization assays are thought to be the most informative surrogate measure for measuring effective immune evasion by influenza virus ^{85–87}.

Over the past several years, we have seen an interesting battle for dominance within the H1N1 A5a clade. The clade split into A5a.1 and A5a.2 subclades in the

2019-2020 influenza seasons. The A5a.1 subclade initially seemed more fit making it appear as though the A5a.2 would be completely replaced by A5a.1. However, the A5a.2 subclade was able to acquire several mutations in its HA protein that caused it not only to survive but thrive and further evolve into two subclades A5a.2a and A5a.2a.1. These new subclades are defined by unique antigenic and receptor binding site mutations including, but not limited to, A186T, Q189E, and E224A in the case of A5a.2a. These mutations have been observed in different countries and Q189E is present in the A5a.1 subclade. A5a.2a.1 has additional mutations including P137S and K142R. Initial serological data from the WHO indicates that while the A5a.2a subclade was already antigenically drifted from the 2022-2023 NH vaccine strain, the A5a.2a.1 was even further removed from the vaccine strain. This information along with the location of these mutations warranted further investigation to better inform vaccine policies. However, this season is also important as it can allow us to decipher the implications of the recent COVID-19 pandemic on influenza spread ^{54,60–62,88}.

4.2 Results

4.2.1 Evolutionary Dynamics of Influenza Subtype H1N1 during the 2022-2023 Season: Diversification of the A5a.2 Subclade.

The 2022-2023 Influenza season was the H3N2 dominant season with the next highest proportion of viruses being from H1N1 followed by a small proportion of viruses from IBV **[Figure 4.1 A]**. H1N1 is constantly evolving into various subclades. Zooming into the 2020-2022 season reveals that while it initially seemed like the A5a.1 subclade was more fit and poised to become the dominant

subclade, the A5a.2 clade not only began to increase in dominance but also increased in diversity, further diversifying into the novel A5a.2a and A5a.2a.1 subclades [Figure 4.1 B].

A Influenza Positive Tests Reported to CDC by U.S. Public Health Laboratories, National Summary, October 2, 2022 – January 21, 2023

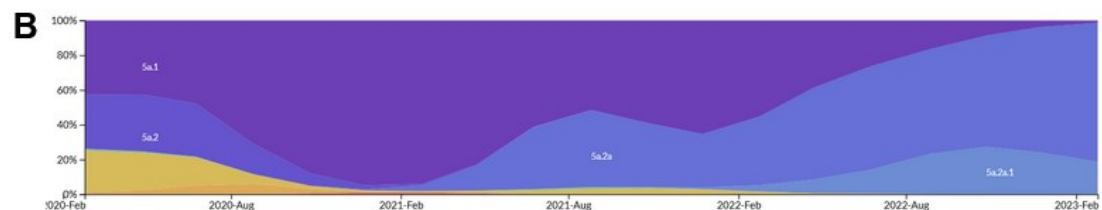
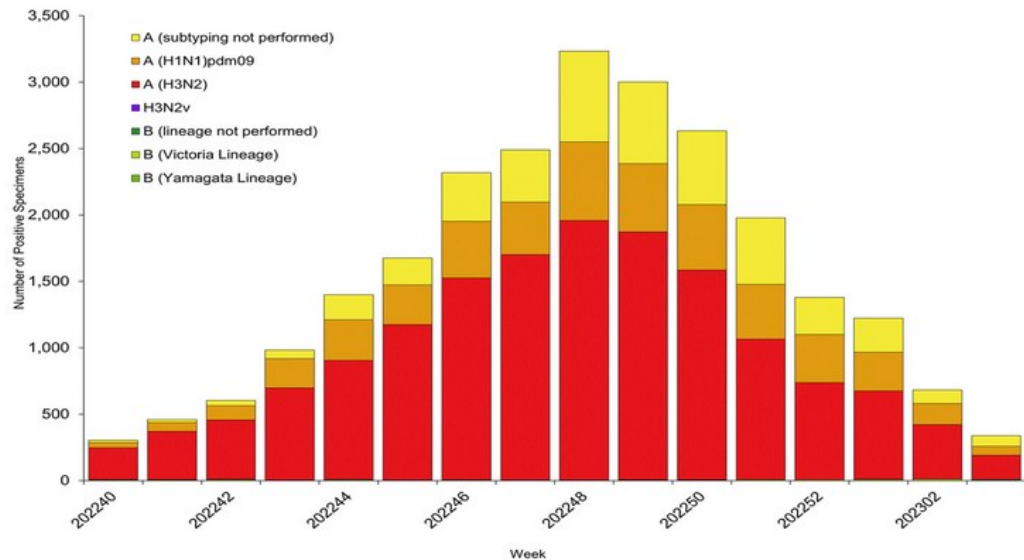


Figure 4.1: Circulation and Emergence of the H1N1 A5a.2, A5a.2a & A5a.2a.1 clades. A) Ratios of Circulating Influenza Viruses in the 2022-2023 season. B) Frequencies of H1N1 clades from Jan 2017 - Feb 2023 generated through NextStrain pipeline.

4.2.2 Representative viruses from A5a, A5a.2, A5a.2a and A5a.2a.1 chosen for characterization.

To study these new subclades and understand the contribution of the mutations accrued, clinical samples sequenced in the Clinical Virology Laboratory of the Johns Hopkins Hospital Network from the 2022-2023 Northern Hemisphere

Influenza season were used for phylogenetic analysis. H1N1 virus sequences from Baltimore were organized into a phylogenetic tree based on their HA sequence using 2019 A5a viruses (A/Baltimore/R0675/2019) as a reference [Figure 4.2]. Representative viruses from each subclade. For further phenotypic characterization, two viruses were chosen from the clade A5a.2a (A/Baltimore/JH-22363/2022 & A/Baltimore/JH-22377/2022) and one virus from A5a.2a.1 (A/Baltimore/JH-22400/2022) [Figure 4.2 B].

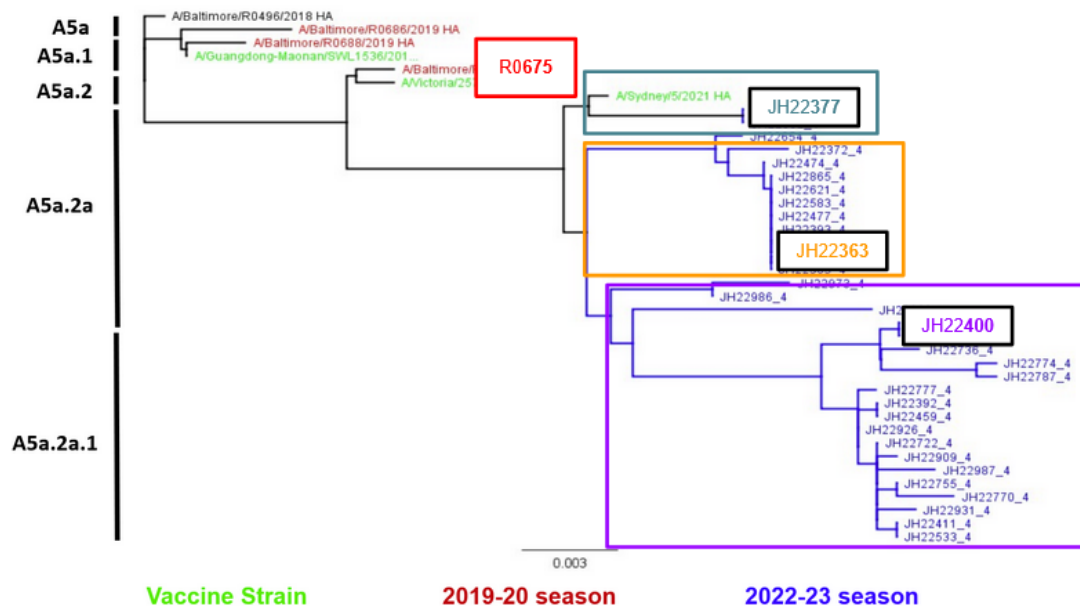


Figure 4.2: Genetic diversity in H1N1 Circulating clades Locally 2022-2023. HA phylogenetic tree showing the different H1N1 HA clades circulating Baltimore during the 2022-2023 Influenza season. 2019 viruses from the A5a.2 clade have been included to indicate genetic differences. Reference strains are in red. The tree was constructed using the Neighbor-Joining method, Tamura-Nei distance model. Viruses chosen for characterization boxed in representative colors.

4.2.3 2022-2023 Circulating H1N1 viruses display improved *in vitro* fitness

Replication differences between the representative viruses were evaluated based on MDCK plaque assays and hNECs growth curves, the latter to recapitulate the upper respiratory tract environment. A low MOI growth curve revealed significant replication differences with both the A5a.2a (A/Baltimore/JH-22363/2022 & A/Baltimore/JH-22377/2022) viruses replicating better than the parental A5a.2 (A/Baltimore/R0675/2019). There were no significant infectious virus titer differences between the parental A5a.2 and the A5a.2a.1 (A/Baltimore/JH-22400/2022) clades **[Figure 4.3 A]**. Plaque assays demonstrated that all the H1N1 viruses circulating in Baltimore during the 2022-2023 influenza season had larger plaque sizes than the parental A5a.2 viruses. There were also differences in plaque sizes between the 5a.2a and 5a.2a.1 viruses **[Figure 4.3 B & C]**.

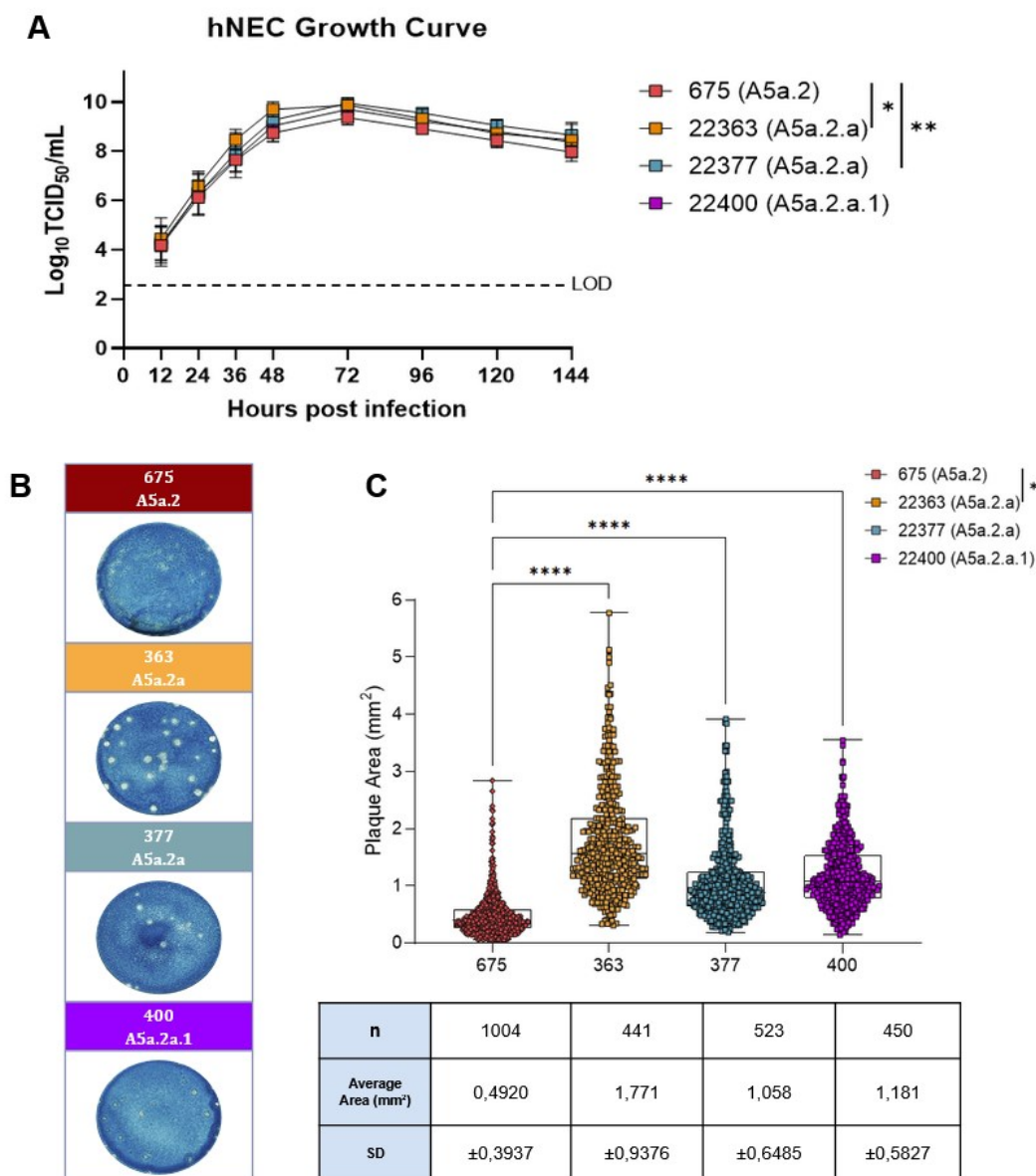


Figure 4.3: *In Vitro* Viral Fitness of 2022-2023 H1N1 Isolates. A) Infectious virus production after infection of hNEC cultures. Data analyzed by two-way ANOVA with Tukey post hoc test. Each graph combines two experiments with three replicates per isolate per experiment * $p < 0.05$, ** $p < 0.01$. B) Representative images of plaques in MDCK cells. C) Quantitation of plaque areas of isolates. Three independent assays with >50 plaques quantified per isolate. One-Way Anova with Tukey post-hoc test, **** $p < 0.0001$ [Growth curve generated by Nicholas Swanson].

4.2.4 All 2022-2023 circulating H1N1 viruses were antigenically drifted from the vaccine

While growth curves and plaque assays can give us an insight into the general fitness of a virus, serum neutralization assays can reveal the implications of antigenic drift on escape from preexisting antibody-mediated neutralizing immunity. Serum neutralization assays were performed with serum collected from Johns Hopkins healthcare workers pre- and post-receipt of the NH 2022-2023 season influenza vaccination.

Table 1: Neutralization Assay Serum Demographics

Serum Cohort	
Gender	No. of Participants
Female	69
Male	30
Age Range: 22-75 years	

Neutralization assays revealed significant differences between the vaccine virus and circulating H1N1 strains and within the different circulating subclades. Pre- and post- neutralization assays revealed that both the 5a.2a and the 5a.2a.1 clades were antigenically drifted from the vaccine strain with the 5a.2a.1 clade being additionally drifted. This baseline neutralizing immunity was boosted by vaccination; however, it did not restore it to the level of the vaccine as measured by the fold change being similar across all viruses. Similar rates of seroconversion

were also observed [Figure 4.4].

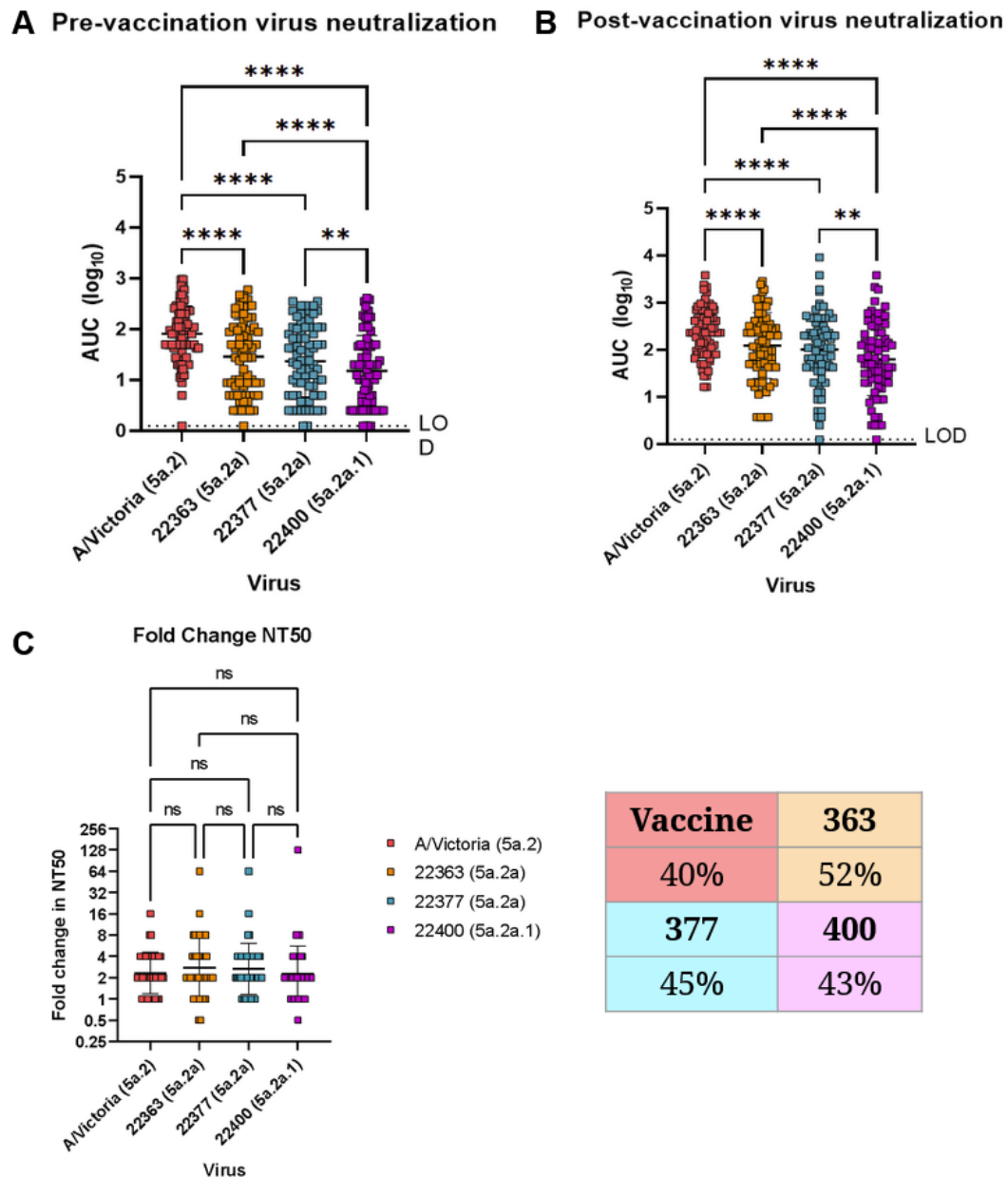


Figure 4.4: Serum neutralizing antibody titers in healthcare workers pre- and post-influenza vaccination. Serum neutralization titers against the H1N1 vaccine strain (A/Victoria/4897/2022), A5a.2a (A/Baltimore/JH22363), A5a.2a (A/Baltimore/JH22377), A5a.2a.1 (A/Baltimore/JH22400) using A) pre- or B) post-vaccination serum. Area under the curve (AUC) was calculated and analyzed using 1-way ANOVA with Tukey post-hoc, **p<0.01, **p<0.0001. C) The fold-change of serum neutralizing antibodies compared across vaccine and circulating isolates and the number of individuals showing a greater than 4-fold rise in neutralization titer. Bars represent the geometric mean; intervals are geometric standard**

deviation. One-way ANOVA with Tukey post-hoc. N=77.

4.2.5 Impact of A5a.2a and A5a.2a.1 mutations on *in vitro* fitness.

To determine the contribution of clade-defining mutations located on important antigenic and receptor binding sites of the HA protein, recombinant viruses were generated using the A5a.2 genetic background (A/Baltimore/R0675/2019) with specific clade defining mutations of the A5a.2a (A186T, Q189E, and E224A) and A5a.2a.1 (P137S and K142R) subclades **[Figure 4.5]**. To ensure that the cloning process did not alter previously observed phenotypes, a reassorted A5a.2 virus with no changes was created and revealed no replication or plaque size differences when compared with the WT virus [work done by Nico Swanson and reported in his PhD thesis].

hNEC growth curves showed that all the point mutations affected replication significantly when compared with the parental strain and that the only mutation that significantly improved replication on hNECs was Q189E **[Figure 4.5 A]**. Plaque assays showed a similar trend of Q189E being the only mutation that increased plaque size while E224A and P137S significantly reduced plaque sizes **[Figure 4.5 B & C]**. Indicating that Q189E could be compensating for the deleterious effects of these other mutations.

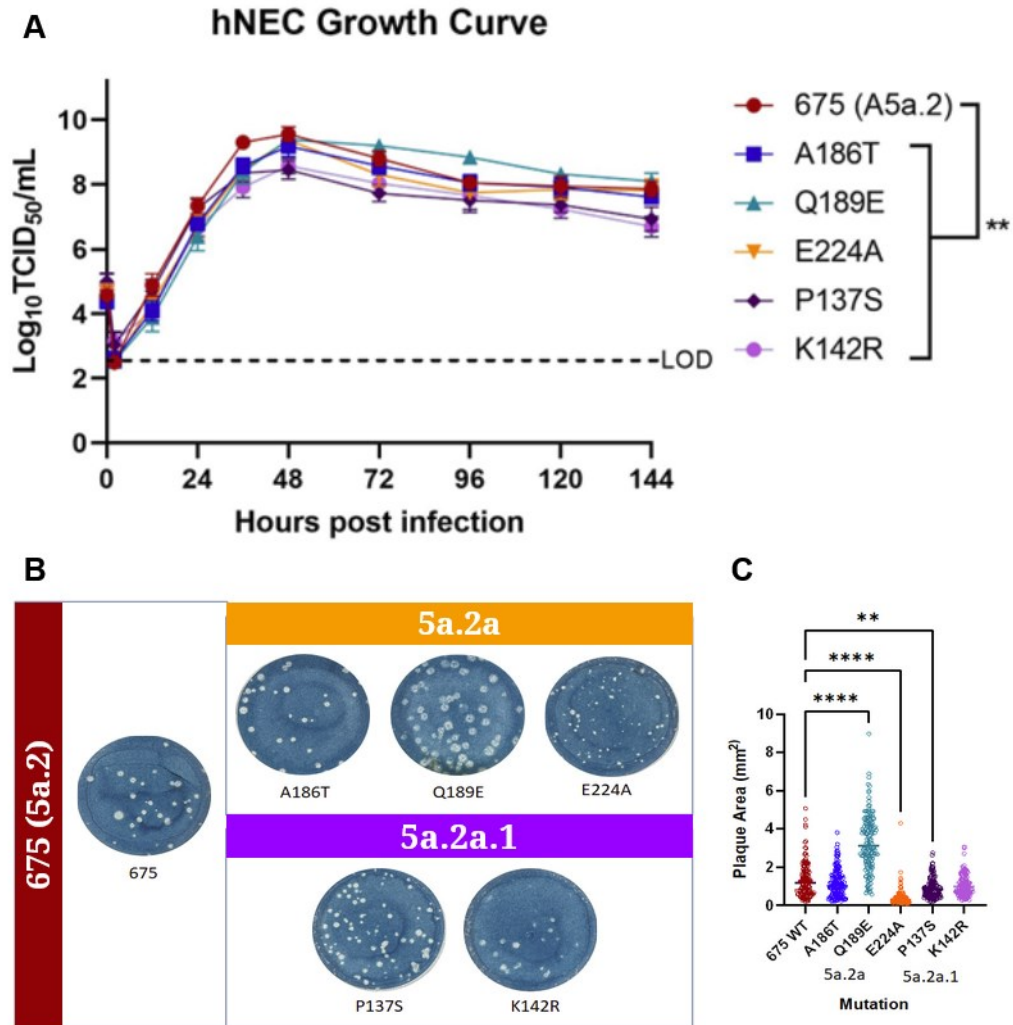


Figure 4.5: In Vitro Virus Fitness of A5a.2a & A5a.2a.1 Mutations in A5a.2 Backbone.
A) Infectious virus production after infection of hNEC cultures. Data analyzed by two-way ANOVA with Tukey post hoc test. Each graph combines two experiments with three replicates per isolate per experiment. ** $p < 0.01$ **B)** Representative images of plaques in MDCK cells. **C)** Quantitation of plaque areas of isolates. Three independent assays with >50 plaques quantified per virus. One-Way Anova with Tukey post-hoc test, ** $p < 0.01$, **** $p < 0.0001$ [Data Generated by Nicholas Swanson].

4.2.6 Q189E compensates for deleterious mutations to a certain extent.

To investigate whether the Q189E compensates for the deleterious mutations (P137S and K142R), plaque assays were performed using WT, point mutant and paired point mutant rescue viruses. As demonstrated in the point mutation plaque assays **[Figure 4.5]**, K142R and P137S reduced plaque sizes while Q189E increased plaque sizes. Combining the two deleterious mutations P137S and K142R resulted in more opaque plaques that resembled the size of P137S alone. Pairing each K142R and P137S with Q189E resulted in larger plaques, however, K142R + Q189E still had more opaque plaques than P137S + Q189E. Putting all three mutations together (as present in the A5a.2a.1 WT 22400 virus) resulted in plaque size and morphologies closer to the WT A5a.2a.1 (22400) virus **[Figure 4.6]**. add a conclusion sentence.

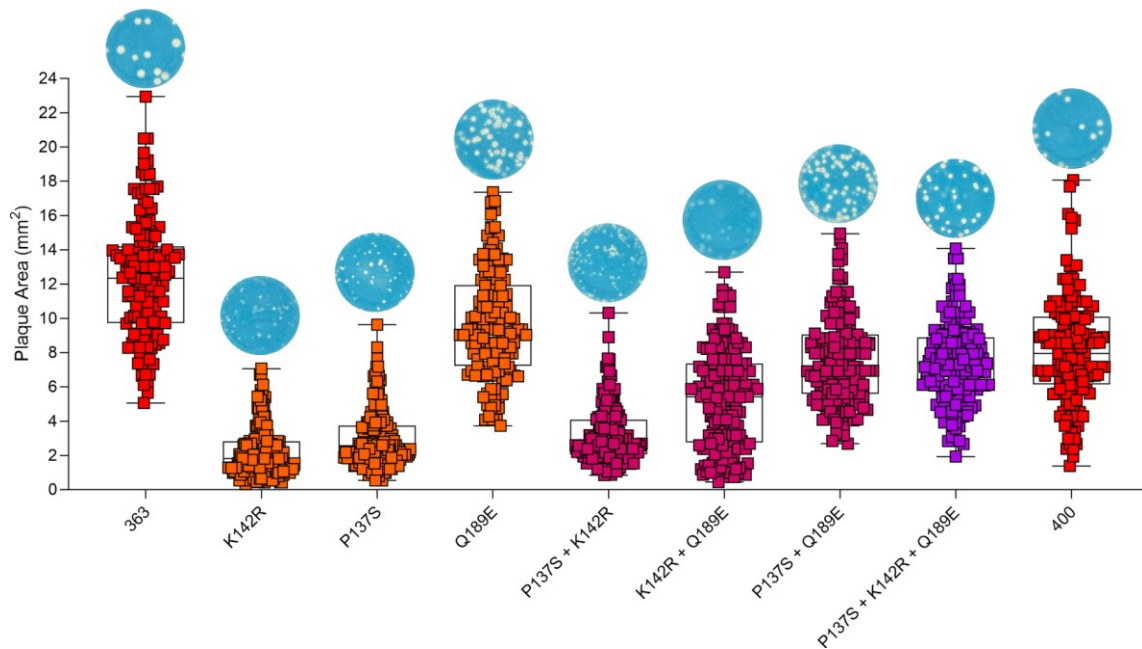


Figure 4.6: Paired point mutant plaque assays. Quantitation of plaque areas of isolates. Three independent assays with >50 plaques quantified per virus. One-Way Anova with Tukey post-hoc test, * $p < .0001$ with representative images of plaques in MDCK cells.**

4.3 Discussion

Influenza is a constantly evolving pathogen that requires updating of vaccine strains to provide maximal protection from severe disease ⁵⁴. That said, the evolution that has occurred within the H1N1 A5a.1 and A5a.2 clades [Figure 4.1B] combined with clues alluding to antigenic drift reducing the vaccine-induced neutralization makes it important to study the contribution of these mutations to inform future vaccine design. *In vitro* assays revealed that the clade defining mutations accrued by the A5a.2a subclade did indeed improve replication fitness, with this subclade replicating significantly better in the physiologically relevant hNEC model and producing larger plaques compared to the parental A5a.2 clade [Figure 4.3]. While the A5a.2a subclade performed better *in vitro* fitness assays,

the 5a.2a.1 subclade seemed to better evade neutralization as displayed by the neutralization assays using serum from healthcare workers **[Figure 4.4]**.

To understand the *in vitro* phenotypes observed, it is important to investigate the mutations that define these clades as well as where they occur **[Figure 4.2]**. While there are many mutations that separate these viruses, the main HA clade defining mutations that define the 5a.2a subclade are A186T, Q189E, and E224A which are present in important regions such as the receptor binding site and important antigenic sites. While 5a.2a.1 has the addition of P137S and K142R. The location of these mutations is consistent with the fitness and antigenic characteristics we have observed through the plaque assays, growth curves and neutralization assays performed (WHO, 2022.) **[Figure 4.3 & 4.4]**.

The clade defining mutations present on antigenic sites all play a role in the fitness and antigenicity of a virus either independently or synergistically^{90–92}. With regards to the neutralization assay results, despite vaccination boosting responses, the Q189E mutation, present in the A5a.1 clade that dominated in the previous season, could have resulted in the 2022-2023 viruses being more prevalent in the population, however, it is possible that while E224A, P137S, and K142R did not improve *in vitro* fitness they might allow the virus to evade antibody-mediated immunity, thus enhancing *in vivo* fitness **[Figure 4.4]**. While this group can offer insights into population immunity, it's crucial to recognize that healthcare workers typically possess greater baseline immunity compared to the general population. This elevated immunity has been linked to reduced rates of seroconversion in the general population, possibly obscuring certain distinctions within this study ^{15,93,94}.

To dissect the contribution of each specific mutation, the infectious clone system^{95–97} was employed to generate viruses with each important mutation on the 5a.2 parental background allowing us to ignore the potential influences of mutations on other segments or other mutations not on antigenic or receptor binding sites. Evaluating these point mutations on the same criteria used to study the A5a.2, A5a.2a, and 5a.2a.1 mutations revealed the true impact of each of these mutations. Interestingly, Q189E was the only mutation that was able to increase plaque size while mutations such as E224A and P137S reduced plaque size. Q189E also allowed the virus to replicate better than all other mutants, specifically after reaching peak titer. All other mutations, on the other hand, seemed to significantly reduce replication. The Q189E mutation occupies the 190 helix of the receptor binding sites and the antigenic site Sb **[Figure 4.5]**. Q189E has now independently appeared in the A5a.1 subclade and the A5a.2a subclades indicating a strong benefit to having this mutation. However, since it has occurred previously and since it is a component of the vaccine this mutation would have to be extremely important for the virus to retain it¹⁵. These findings suggest that while the clade defining mutations other than Q189E negatively impact fitness they help the virus evade the immune system.

To fully investigate the contribution of the Q189E mutation, paired point mutants of different combinations were generated. As previously demonstrated **[Figure 4.5]**, P137S and K142R individually reduce plaque size while Q198E increases plaque size. What is important to note is that apart from the size of plaques the mutations also seem to influence the opacity of the plaque created with K142R

having the opaquest plaques. Combining the mutations that reduce plaque size (P137S + K142R) resulted in plaque sizes that are similar to P137S alone (slightly larger than K142R) that were opaquer. Furthermore, P137S +K142R also took the longest to grow and appeared to have no cytopathic effect during rescue but did produce a TCID50 titer. Indicating some kind of replication cost that could be better captured through hNEC growth curves. While adding Q189E to each P137S and K142R increased the size of plaques, K142R + Q189E plaques were still more opaque in comparison to the P137 +Q189E plaques. Combining P137S and K142R with Q189E resulted in majority clear plaques with sizes similar to the 400 (5a.2a.1) virus with all three of these mutations present. All viruses involving some combination of P137S and/or K142R were smaller than the 363 (5a.2a) WT virus **[Figure 4.6]**. Taken together, this dynamic is representative of the constant struggle between immune evasion and fitness. While the viruses take on the probability of being recognized by maintaining the Q189E mutation that has been present in its predecessors, it seems to be in the virus's interest to keep it to make up for the potentially deleterious other mutations such as P137S and K142R it has acquired that allows it to drift away from the vaccine strain and perhaps even pre-existing immunity.

Following the rise and diversification of the A5a.2 H1N1 clade, Baltimore surveillance, like global surveillance, revealed that viruses were predominantly from the A5a.2a and A5a.2a.1 subclades **[Figure 4.1 & 4.2]**. Fitness assays revealed that all the 2022-2023 viruses produce larger plaques possibly due to the beneficial effects of the Q189E mutation that is making up for the deleterious

mutations that in turn aid in immune evasion **[Figure 4.3 & 4.5]**. This hypothesis is supported by the single point mutation in infectious clones that were paired with Q189E **[Figure 4.6]**. Neutralization assays showed that while vaccination boosted responses, it did not restore it to the level of observed for the vaccine strain. Neutralization assays combined with the fact that all relevant clade defining mutations other than Q189E were detrimental to fitness or did not impact it means that they could be beneficial to immune evasion or increase receptor binding **[Figure 4.4]**.

This combination of mutations that improve fitness or drive antigenic drift mirror the constant fluctuation between the two and can be explained by the concept of epistasis ^{98,99}. Future studies should be geared toward neutralization assays of the single and paired point mutants and the same set of serum to confirm that the mutations that do not benefit fitness, benefit the virus through immune evasion. Growth curves with the paired point mutations can further validate the contribution of Q189E. Mutations can also improve receptor binding¹⁰⁰ or binding avidity, therefore, glycan arrays and avidity assays can provide additional insight into receptor binding changes ^{101–103}.

4.4 Chapter Specific: Materials and Methods

4.4.1 Cell Lines

As described in the methods section **[Chapter II]** with the addition of human embryonic kidney cells 293T (HEK293T).

4.4.2 Growth Curve, Plaque Assay, Neutralization Assay & Statistical Analysis

As described in the methods section [**Chapter II**]. With the following viruses:

- A/Baltimore/R0675/2019
- A/Baltimore/JH-22363/2022
- A/Baltimore/JH-22377/2022
- A/Baltimore/JH-22400/2022
- A/Baltimore/R0675/2019 with **A186T** [Plaque Assay and Growth Curve Only]
- A/Baltimore/R0675/2019 with **Q189E** [Plaque Assay and Growth Curve Only]
- A/Baltimore/R0675/2019 with **E224A** [Plaque Assay and Growth Curve Only]
- A/Baltimore/R0675/2019 with **P137S** [Plaque Assay and Growth Curve Only]
- A/Baltimore/R0675/2019 with **K142R** [Plaque Assay and Growth Curve Only]
- A/Baltimore/R0675/2019 with **P137S + Q189E** [Plaque Assay Only]
- A/Baltimore/R0675/2019 with **K142R + Q189E** [Plaque Assay Only]
- A/Baltimore/R0675/2019 with **P137S + K142R** [Plaque Assay Only]
- A/Baltimore/R0675/2019 with **P137S + K142R + Q189E** [Plaque Assay Only]

4.4.3 Plasmids

H1N1 A5a.2 rescue viruses were generated by using a pHH21 plasmid backbone which has a DNA sequence for the BsmBI restriction site which, after restriction enzyme digestion, reveals a palindromic nucleotide sequence that is utilized for the ligation to DNA digested by BsmBI or other restriction sites such as BsaI. Viral RNA was extracted with the QIAamp viral RNA mini extraction kit and converted into cDNA with universal influenza A primers. Each segment was then amplified from the cDNA with Phusion High fidelity DNA polymerase (New England Biolabs) with segment specific primers possessing restriction enzyme sites at the ends of the segments. All segments and the plasmid were digested using restriction enzymes (such as BsmBI or BsaI, New England BioLabs) and ligated together using the T4 DNA ligase (New England BioLabs). Ligated plasmids were then transformed into a DH5 α bacteria (Thermo Fisher) and cultured overnight on LB

plates with carbenicillin to allow for the selection of colonies that have incorporated the plasmid. Individual colonies were picked and cultured for plasmid extraction. Plasmid extraction was done using a Qiagen QIAprep Spin Miniprep Kit. Sequences of the plasmid were verified through restriction enzyme digestion and whole plasmid sequencing through Plasmidsaurus using Oxford Nanopore Technology. Bacteria with the desired plasmid was then grown in bulk and extracted through the ZymoPURE Plasmid Midiprep kit (Zymo research) and verified again through whole plasmid sequencing. Single point mutations were generated by mutating single nucleotides on primers designed for specific locations through New England BioLabs software NEBaseChanger. Primers were used on the A/Baltimore/R0675/2019 HA to produce plasmids with the required mutations. PCR was conducted using the Q5 site-directed mutagenesis kit (New England Biolabs). Transformation, extraction, and confirmation of sequences were the same as described^{104–106}.

4.4.4 Recombinant Virus Generation

Recombinant viruses were generated using a 12-plasmid reverse genetics system^{105,107–109}. To each well in a six well plate 500 μ L of poly-L-lysine was added and rocked to allow even distribution. All wells were then washed two times with PBS+. HEK293T cells were seeded at a confluency of 2×10^5 cells/well. The plate was incubated at 37°C for 20-24 hours. After the cells reached a confluency of between 40-60%, the medium was changed to serum free Opti-MEM (Thermo Fisher Scientific). TransIT-LT1 (Mirus Bio) is then vortexed at room temperature and used to make a master mix with Opti-MEM (Thermo Fisher Scientific). Each of the 8 segment plasmids (PB2, PB1, PA, HA, NP, NA, M, NS) were then combined with aliquoted master mix. After adding all plasmids at the required concentrations, the plate was incubated at 32°C for 24 hours and overlaid with MDCK cells. Plates were then incubated again at 32°C and 1 mL of supernatant is collected every

24 hours and replaced with 1 ml of freshly prepared infection media [As described in Chapter 2, Section 2.3] per well. The collected supernatant was then tittered using the previously described TCID₅₀ protocol [As described in Chapter 2, Section 2.4] and the supernatants of the wells with the lowest concentrations of virus were collected to grow stocks.

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